Likelihood approximations distort the ion channel kinetic information in macroscopic currents

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Abstract Kinetic schemes are usually fitted to macroscopic ion-channel currents by least squares on the deterministic mean, leaving the gating fluctuations unmodelled. Likelihoods that use them, including Bayesian recursive filters, are two decades old and little used: a filter tracks the data even when the uncertainty it reports is wrong. The test is classical: at the true parameters the score, the log-likelihood's derivative, averages to zero, and its covariance across recordings equals the Fisher information it reports. We measure bias and distortion along a seven-rung ladder from least squares to a filter conditioned on both ends of each acquisition interval, for a two-state scheme over channel number, noise and acquisition interval, against exact simulations of the process they approximate. The boundary-conditioned filter is calibrated over most of the sweep, departing toward few channels and low noise, where the occupancy Gaussian is misspecified. Least squares becomes calibrated once instrumental noise exceeds gating noise, which happens at an instrumental variance proportional to channel number. No measured recording gives both the unitary current and a trustworthy least-squares error bar, the two boundaries one to three decades of noise apart. The boundary-conditioned likelihood (MacroIR) is available for R and Python with its score and information, so an experimenter can fit their own recording.

Introduction

A channel opens and shuts at random, and how long it stays in each conformation is the elementary random quantity of gating. What an experiment sees of it depends on how many channels contribute. With one channel in the patch the durations are read off the record and the rate constants come from their distributions (***Neher and Stevens, 1977***; ***Colquhoun and Hawkes, 1981***). In the currents that carry an action potential so many channels contribute that the durations average away, the current follows the deterministic relaxation of the Hodgkin and Huxley description (***Hodgkin and Huxley, 1952***), and rate constants are obtained by fitting that relaxation, in current practice by least squares.

Most macroscopic patch-clamp recordings are in neither place. A few hundred to a few thousand channels contribute, individual openings cannot be resolved, and the averaging is incomplete. The durations are still there, and what they leave behind is a fluctuation around the mean current that carries the memory of the gating process. Samples taken closer together than the channel's relaxation time are correlated, so a sample taken faster than that time partly repeats what the one before it already said.

That fluctuation carries what the mean cannot. Channel number and unitary current enter the mean current as a product, so no fit to the mean separates them. The standard analysis discards the fluctuation anyway: the four approaches surveyed by ***Clerx et al.*** (***2019***) are deterministic fits with a root-mean-square objective, and the IonBench benchmark states that it does not cover optimisation approaches for stochastic models (***Owen and Mirams, 2025***). Discarding it usually leaves the point estimate standing. What it damages is the count of independent residuals, the number that turns a residual sum of squares into a confidence interval, so a fit that treats correlated residuals as independent behaves as though the recording had supplied more observations than it did. ***Del Core and Mirams*** (***2025***) state the same cost in the field's own terms, that standard methods built for deterministic models do not distinguish stochastic channel gating from measurement noise and return biased estimates.

A Markov description models the dependence rather than assuming it away. A channel with a small number of conformational states, connected by rate constants that hold over the whole record, generates the mean current and the covariance of the fluctuations from the same constants, so the temporal structure of the noise becomes a second measurement of the same parameters. Those constants also mean something outside the record. They are properties of the protein, comparable across conditions, preparations and laboratories, and they connect what is measured to a mechanism. A correlation function fitted freely to the residuals delivers neither of those.

Kinetic information has been read from that correlation since the 1970s, first through stationary power spectra (***Katz and Miledi, 1970***, ***1972***; ***Anderson and Stevens, 1973***), from which ***Anderson and Stevens*** (***1973***) obtained a unitary conductance and a closing rate jointly with standard errors on both, and then through the exact non-stationary two-time covariance of a population of identical independent channels, estimated from ensembles of 256 to 504 repeated sweeps (***Conti et al., 1980***; ***Sigworth, 1981***). The whole-record likelihoods that followed differ along two axes and are best read that way. The first is how much of the gating stochasticity the likelihood models: nothing, then the variance at each sample (***Milescu et al., 2005***), then its correlation in time, whether by carrying the full two-time covariance at a cost that keeps it to a hundred or two selected points (***Celentano and Hawkes, 2004***), or by propagating the occupancy recursively and conditioning it on each new datum at a cost linear in the record (***Moffatt, 2007***; ***Stepanyuk et al., 2011***; ***Münch et al., 2022***). The second axis is the observable itself. An acquisition system reports an average over each interval rather than an instantaneous sample, and the most recent member conditions that interval average on the channel states at both of its ends (***Moffatt and Pierdominici-Sottile, 2025***).

None of these has displaced least squares. Cost is part of it, and ***Del Core and Mirams*** (***2025***) make that case. There is also something these methods ask an analyst to give up. A least-squares fit predicts the current at every point from the parameters alone, so the residual can be read directly, and read twice. Its amplitude reports the quality of the fit against what the gating variance and the baseline noise should deliver, and its correlation in time reports what the fit leaves out. Both readings appeal to knowledge from outside the fit, which is what makes them checks.

Modelling the correlation is what takes those readings away. A recursive likelihood predicts each interval from the parameters and from the data already observed, so the previous sample enters the prediction of the next one and the predicted trace follows the record by construction. It follows the record when the scheme is right, when the scheme is wrong, and when the code has a mistake in it. The limiting case of the argument carries no kinetics at all and states that each sample equals the last one plus noise: where the interval is short against the relaxation time and the instrumental noise is small, it predicts almost perfectly and says nothing about the channel. Such a likelihood also standardises its residual against its own predicted variance, so the amplitude reading becomes a statement of internal consistency, and the correlation reading is spent because the recursion subtracts at each step what it has just seen. The same blindness covers two questions that an experimenter cannot separate and has no reason to: whether the approximation is adequate, and whether the code that computes it is correct.

The apparatus that answers this is classical. For a correctly specified likelihood the score, the gradient of the log-likelihood with respect to the parameters, has mean zero at the parameters that generated the data, and its covariance across independent recordings equals the Fisher information the likelihood reports (***Huber, 1967***; ***White, 1982***). An approximation satisfies neither statement at those parameters, and the two departures are measurable quantities rather than tests to be passed. Goodness of fit does not answer the question, since an approximation can reproduce the mean current closely and still report an information matrix wrong by more than an order of magnitude, and comparing two approximations against each other does not answer it either, because both may be wrong. Measuring the departures takes an ensemble of recordings drawn from known parameters, which is why this is a simulation study and not something a reader can run on a single record. What makes it possible at all for this class of model is an asymmetry between simulation and inference: the forward process can be simulated exactly while its likelihood cannot be computed exactly, so the simulator supplies ground truth for what the likelihood approximates.

Two alternatives to a mechanistic likelihood are worth placing here, since both are in use. The standard statistical repair for correlated residuals is generalized least squares with autoregressive or moving-average errors, which whitens the residual sequence with a correlation function estimated from the residuals themselves, and ***Lei et al.*** (***2020***) have run exactly that on ion-channel data. Gating puts that family out of reach for structural reasons. An autoregressive moving-average error model is stationary by construction, one covariance function for the whole trace, while the gating covariance tracks the mean current, vanishes when the current does, and restarts at every concentration or voltage jump. It contains no channel number and no unitary current, so it cannot return the two parameters the fluctuations are richest in. Its timescales are free parameters, where the Markov model ties them to the eigenvalues of the rate matrix that already fixed the mean, so free timescales absorb the information the mechanism would have contributed. ***Lei et al.*** (***2020***) report that the autoregressive model widens the posterior without predicting better. The other alternative is non-stationary fluctuation analysis, the one fluctuation-based method in routine use, which regresses the variance of the current against its mean and returns the unitary current, the channel number and the peak open probability (***Stepanyuk et al., 2014***). This paper does not compete with it over how much amplitude information a macroscopic recording holds, never benchmarks against it, and no comparison with the variance-mean regression should be read into any result below.

Because every member is an approximation, the measurement returns a domain rather than a verdict. The question is where a likelihood reports honestly, and the answer is a map whose boundaries are level sets of a continuous diagnostic rather than thresholds. Position on it is set by the number of channels, by the acquisition interval in units of the channel's relaxation time, and by the instrumental noise in units of the gating noise, held at fixed spectral density so that shortening the interval raises the noise in each sample and leaves the noise per unit bandwidth unchanged. The measurements are made on a two-state scheme at an open probability of one half, driven by a single concentration jump, compared at the level of the likelihood rather than of a fitted experiment. The small model isolates the error the approximation itself commits, with no confounding from a misspecified kinetic scheme, it keeps the exact simulation cheap enough to run thousands of replicates at every design cell, and it is the most favourable case for a Gaussian treatment of the occupancy and therefore a lower bound on the error for anything richer. Open probability, stationary protocols and larger schemes are left out for reasons that follow from the same design, and are stated with the results.

Two things come out of the map, and they answer different readers. The first is where each approximation misreports what the data determine, including the most complete one, since a method whose limits are unmapped has been endorsed rather than characterised. The second is the region in which a classical fit reports an interval that matches the spread it delivers, and it carries an ordering that is falsifiable and does not depend on where the level sets are placed: there is no region of the measured plane in which the fluctuations still carry information about the unitary current or the channel number and the classical interval is simultaneously trustworthy, the two boundaries being separated by one to three decades of instrumental noise everywhere they were measured.

Against that history, what is new here is narrow and is worth stating narrowly. A joint estimate of rates and amplitudes from fluctuation data, carrying standard errors, is ***Anderson and Stevens*** (***1973***). What this paper adds, to a non-stationary protocol read from a single record, is the third element of the combination: an interval whose calibration has been verified rather than assumed. The thousands of replicate recordings used below certify the method; they are not what applying it requires. Separately from the map, there is a reading an experimenter can take from a recording they already have. The integrated autocorrelation of the standardized least-squares residual counts how many intervals the fit effectively has for every interval it believes it has, and it costs nothing, since the fit was going to be made anyway. Two design variables that look as though they should matter, the channel count and the instrumental noise, enter the answer through that one statistic and through nothing else. The acquisition interval does not, and the reader knows it, because they chose it when they set the sampling rate. Those two numbers together say what a classical fit is doing to the width of the interval it reports, in units a reader can act on: a distortion of 1.15 is a 7% error on a confidence interval, 2 is 41%, and 4 turns a nominal 95% interval into about 68%.

The macroscopic interval likelihood and its approximations

A macroscopic current is the summed signal of a population of ion channels, each of which behaves as an independent continuous-time Markov chain over a small set of conformational states. We write N_{ch} for the number of channels and K for the number of kinetic states of one channel ($K = 2$ throughout this paper, carried symbolically where it costs nothing). Each channel is governed by a generator (rate) matrix $\mathbf{Q} \in \mathbb{R}^{K\times K}$, whose transition matrix over an elapsed time t is the row-stochastic $\mathbf{P}(t) = e^{\mathbf{Q}t}$ with entries $P_{i\to j}(t)$. A channel in state k carries a conductance γ_k, and we collect these in the vector $\boldsymbol{\gamma} \in \mathbb{R}^K$.

The likelihood we need is the probability of a recorded sample given the kinetic parameters and the data seen so far. Its exact form would require the distribution of the interval-averaged conductance of the whole population, which has no closed expression. Every practical likelihood replaces that distribution with a Gaussian, and the members of the family studied here differ in what the Gaussian is conditioned on and in how its variance is accounted. The rest of this section unpacks that sentence.

The observable is an interval average

A recorded sample is the current averaged over the acquisition window rather than sampled at an instant. The current is low-pass filtered in analogue form before it reaches the converter, so each stored value carries a weighted average of the recent current and not a reading of it at a point. Writing the sampling period as Δ and taking the weighting to be uniform across it, the quantity a likelihood must describe is the windowed average

$$\bar{y}_t \;=\; \frac{1}{\Delta}\int_{t-\Delta}^{t} y(\tau)\,\mathrm{d}\tau \;+\; \eta_t, \tag{1}$$

where $y(\tau)$ is the instantaneous population current and η_t is the instrumental noise accumulated over the window. This interval average is the physical observable. When Δ is much shorter than the fastest relaxation of $\mathbf{Q}$ the average sits close to the instantaneous current and the distinction is negligible; when Δ is comparable to or longer than that relaxation the two diverge, and a likelihood built on instantaneous sampling describes the wrong random variable.

Uniform weighting is a first approximation to what the electronics do. The analogue stage of a patch-clamp amplifier is normally a Bessel low-pass, chosen because its group delay is nearly flat and its step response barely overshoots, so the effective measurement kernel is that filter's impulse response, a smooth shape whose tails reach into the neighbouring intervals. The uniform window keeps the entire kernel inside one acquisition interval, which is what lets a likelihood conditioned on the two ends of that interval account for the averaging without reference to anything outside it. A Bessel kernel breaks that containment twice over: it correlates successive recorded values through the signal, and it colours the instrumental noise that Eq. 1 takes to be white. We keep the uniform window throughout, in the simulator as well as in every likelihood (Methods), so that data and likelihood describe the same observable and the closures below are the approximation the diagnostic sees. The uniform window is itself an approximation to the impulse response of the anti-aliasing filter of a real recording, and the Discussion prices it; it does not enter the measurement here because the simulator and every member share it. kernel, with the coloured noise that comes with it, through the same construction is left to later work.

Two Gaussian approximations

Each member of the family makes two closures, on two different objects. To close a distribution is to replace it by the Gaussian with the same mean and covariance, discarding every higher moment. The word is borrowed from moment hierarchies, where that substitution lets a finite set of equations stand for an infinite one, and it fits here for the same reason: conditioning the occupancies on an observation takes them out of the multinomial family, and the exact conditional distribution after t samples has no summary of fixed size, whereas $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is the same size at every step.

The two Gaussian approximations

Occupancy closure. The joint distribution of the channel occupancies is exactly multinomial when the channels are independent, which is also the maximum-entropy distribution consistent with the mean occupancies. It is replaced by a multivariate Gaussian with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, valid for large N_{ch} by the central limit theorem and degrading as the channel count falls.

Interval-signal closure. The distribution of the interval-averaged conductance over one window is a telegraph-like object with no closed form. It is replaced by a Gaussian, valid when the window spans many transitions or when instrumental noise dominates, and degrading as the window shrinks below the relaxation time.

The higher moments these closures discard do not vanish. They reappear in the data as temporal correlation that the likelihood does not predict, which is what the correlation term of the information-distortion decomposition measures (Diagnostics). Three regimes follow, and we name them here so the maps in the Results have vocabulary: the *multinomial* regime (few channels, where the occupancy closure fails), the *telegraphic* regime (windows far shorter than the relaxation time, where the interval-signal closure fails), and the *Gaussian* regime, where both closures hold because the channel count is high enough for the occupancy Gaussian while the window is long enough, or the instrumental noise large enough, for the interval-signal Gaussian. Any Gaussian likelihood of this kind must fail somewhere. The aim of this paper is to show that the failures are the predicted degradation of these two closures rather than defects of a particular algorithm.

The usage map of the Results is drawn on the plane of channel count against instrumental noise at one acquisition interval, and its region names follow this vocabulary. Region 0, where the closure itself is misspecified, is the multinomial regime seen on that plane, and its boundary carries a negative slope because instrumental noise makes the emission more Gaussian and so relaxes the occupancy closure. Regions 1 to 4 (the unitary current and channel count with a calibrated interval; the rates with a calibrated interval; least squares at par; nothing estimable) all sit inside the Gaussian regime, and what orders them is how much the data carry rather than whether the closures hold. The telegraphic regime is a statement about Δ and so does not appear as a region of that plane; it is reached by moving along the interval axis of Figure 4.

The root question, and the ladder below it

Before any of the choices above comes a prior question: does the method model the gating fluctuations at all? Classical nonlinear least squares (`LSE`) answers no. It fits the deterministic mean current and assigns one constant variance to every residual, so it carries neither an occupancy distribution nor a conductance conditioned on anything, and the temporal structure the two closures approximate is absent from it. `LSE` is therefore not a member of the family. It is the bottom rung of the ladder of computational cost this paper walks, and it is the anchor against which every member above it is measured, because it is the method most readers are using.

Given a yes to the root question, the members differ along two axes. One axis is the window: how much of the acquisition interval the conductance is conditioned on. The other is recursion: whether the occupancy distribution is conditioned on the data as the recording proceeds.

Along the recursion axis, a recursive (prequential) member factorizes the likelihood as $\prod_t q_t(\bar{y}_t \mid \bar{y}_{1:t-1})$, so the occupancy distribution carried into each window is the posterior left by the previous windows and the hidden state is conditioned on the data. A non-recursive (open-loop) member factorizes as $\prod_t m_t(\bar{y}_t)$, with each window's predictive propagated from the initial condition and never conditioned on the observed trace.

Along the window axis the members form a ladder, indexed by how many of the interval's endpoints the conductance is conditioned on (Table 1).

Conditioned on	Members	Conductance model
the fluctuations are not modelled	`LSE`	the deterministic mean current only
no endpoints	`NR`, `R`	instantaneous; the acquisition averaging is ignored
one endpoint (the start)	`MR`, `VR`	interval-mean conductance given the start state
endpoints not distinguishable	`INR`	interval-mean conductance; with no update the two endpoint rows coincide
two endpoints (the boundary)	`IR`	interval-mean conductance given both boundary states; interior marginalized
the full trajectory	(exact)	intractable; supplied by the stochastic simulation as ground truth

Table 1. The window axis as an endpoint ladder, with least squares (`LSE`) shown at the foot of the cost ladder although it lies off the lattice. Crossing the window axis with the recursion axis gives the implemented grid: `NR` (non-recursive, $\mathrm{av} = 0$), `INR` (non-recursive, $\mathrm{av} = 1$, total start-conditioned interval variance), `R` (recursive, $\mathrm{av} = 0$), `MR` (recursive, $\mathrm{av} = 1$, total start-conditioned interval variance), `VR` (recursive, $\mathrm{av} = 1$, residual interval variance), `IR` (recursive, $\mathrm{av} = 2$), where av counts the conditioned endpoints. `INR` occupies a row of its own because without a recursive update the boundary state is unobservable: the two-endpoint conductance can enter only through its start-state marginal, so $\mathrm{av} = 1$ and $\mathrm{av} = 2$ describe the same non-recursive method. The body walks `LSE`, `NR`, `R`, `IR`. Of the three reported in the Supplement, `MR` and `VR` split the step from `R` to `IR`. MacroIR is `IR` and MacroINR is `INR`.

All seven rungs are implemented and measured, and Figure 3 carries them side by side under one set of conditions, so the lattice is read as a design rather than as a list.

Two things follow from the ladder. First, `IR` occupies the top rung below the intractable exact likelihood, so its standing in the family is structural and set before any measurement. Second, the lower rungs are established likelihoods rather than constructed foils: the non-recursive instantaneous member is the independent-interval likelihood of ***Milescu et al.*** (***2005***), and the recursive instantaneous member is the macroscopic recursive filter of ***Moffatt*** (***2007***) and the generalized filter of ***Münch et al.*** (***2022***). The lattice is one coordinate system that contains these methods as special cases.

The boundary state

A *boundary state* is the pair (i_0, i_t) of a channel's states at the two ends of an acquisition interval. Chaining intervals, the end state of one interval is the start state of the next, so there is a single state variable per junction: the filter conditions on the states at the interval boundaries and marginalizes the trajectory between them analytically. This is the same device as static condensation, or the spatial Markov property, applied on the time axis. It is not a transition state in the mechanistic sense, and the word *transition* is not used for it here. Because the joint dynamics of the two endpoints are generated by the Kronecker sum $\mathbf{Q} \oplus \mathbf{Q} = \mathbf{Q} \otimes \mathbf{I} + \mathbf{I} \otimes \mathbf{Q}$, whose propagator factorizes as $e^{(\mathbf{Q}\oplus\mathbf{Q})t} = e^{\mathbf{Q}t} \otimes e^{\mathbf{Q}t}$, the moments over the K^2 boundary pairs collapse to bilinear forms in the K-dimensional state space and no explicit K^2 object is formed (Methods).

The ladder needs two conductance objects, both built from the boundary-conditioned single-channel moments $\overline{\Gamma}_{i_0\to i_t}$ (the mean interval-averaged conductance of a channel that starts in i_0 and ends in i_t) and $\overline{V}_{i_0\to i_t}$ (its variance). Both are moments of the integrated conductance of a single channel over one window,

$$A(\Delta) = \int_0^\Delta \gamma\big(X_s\big)\,\mathrm{d}s, \tag{2}$$

taken conditional on the chain starting in i_0 and ending in i_t, and both are available in closed form without diagonalizing $\mathbf{Q}$ and without forming any K^2 object. Write $\mathbf{\Gamma} = \mathrm{diag}(\boldsymbol{\gamma})$ and build the two augmented generators

$$\mathcal{Q}_2 = \begin{pmatrix} \mathbf{Q} & \mathbf{\Gamma} \\ \mathbf{0} & \mathbf{Q} \end{pmatrix}, \qquad \mathcal{Q}_3 = \begin{pmatrix} \mathbf{Q} & \mathbf{\Gamma} & \mathbf{0} \\ \mathbf{0} & \mathbf{Q} & \mathbf{\Gamma} \\ \mathbf{0} & \mathbf{0} & \mathbf{Q} \end{pmatrix}, \tag{3}$$

of sizes $2K \times 2K$ and $3K \times 3K$. Their exponentials are block triangular with $e^{\mathbf{Q}\Delta}$ on the diagonal, and the two blocks above it are the objects wanted: $e^{\mathcal{Q}_2\Delta}$ has $\mathbf{F}_1(\Delta)$ in its $(1,2)$ block and $e^{\mathcal{Q}_3\Delta}$ has $\mathbf{F}_2(\Delta)$ in its $(1,3)$ block, where $\mathbf{F}_1$ and $\mathbf{F}_2$ are the first and second derivatives of the tilted semigroup $e^{(\mathbf{Q}+\alpha\mathbf{\Gamma})\Delta}$ at $\alpha = 0$, the second up to the factor $\mathbf{W} = 2\mathbf{F}_2$ that comes from the two triangular halves of the double integral. This is the block construction of ***Van Loan*** (***1978***). Dividing by the transition probability conditions on the endpoints and dividing by Δ turns the integral into an average, so

$$\overline{\Gamma}_{i_0\to i_t} = \frac{[\mathbf{F}_1(\Delta)]_{i_0 i_t}}{\Delta\, P_{i_0\to i_t}(\Delta)}, \qquad \overline{V}_{i_0\to i_t} = \frac{2[\mathbf{F}_2(\Delta)]_{i_0 i_t}}{\Delta^2\, P_{i_0\to i_t}(\Delta)} - \overline{\Gamma}^2_{i_0\to i_t}. \tag{4}$$

Both expressions are $0/0$ where a transition is unreachable, which is not a numerical edge case here: at zero agonist $k_{\mathrm{on}}[A]$ vanishes identically, so $P_{C\to O}(\Delta) = 0$ exactly over the whole pre-pulse segment. Conditioning on a transit that occupies a vanishing fraction of the window leaves the chain at its two endpoint states for essentially equal times, so the limit is the endpoint average,

$$\overline{\Gamma}_{i_0\to i_t} \to \tfrac{1}{2}(\gamma_{i_0} + \gamma_{i_t}), \qquad \overline{V}_{i_0\to i_t} \to \left(\tfrac{\gamma_{i_0}-\gamma_{i_t}}{2}\right)^2, \tag{5}$$

which depends on the conductance vector alone and introduces no free constant. The start-conditioned mean conductance is the K-vector

$$(\bar{\gamma}_0)_{i_0} \;=\; \sum_{i_t} P_{i_0\to i_t}(\Delta)\,\overline{\Gamma}_{i_0\to i_t}, \tag{6}$$

which conditions on the state at the interval's start alone and is the object used at $\mathrm{av} = 1$. The boundary-conditioned mean conductance is the same average retained on both endpoints, so it stays a $K \times K$ object rather than collapsing to a vector, and is the object used at $\mathrm{av} = 2$.

The interval update

Over a window of length Δ, the occupancy mean and covariance propagate as

$$\boldsymbol{\mu}^{\mathrm{prop}} = \mathbf{P}(\Delta)^{\top}\boldsymbol{\mu}_0, \tag{7}$$

$$\mathbf{\Sigma}^{\mathrm{prop}} = \mathbf{P}(\Delta)^{\top}\big(\mathbf{\Sigma}_0 - \mathrm{diag}(\boldsymbol{\mu}_0)\big)\mathbf{P}(\Delta) + \mathrm{diag}(\boldsymbol{\mu}^{\mathrm{prop}}), \tag{8}$$

where $(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ is the occupancy Gaussian at the start of the window. Every occupancy moment in this section is *per channel*: with $\mathbf{N}_0$ the vector of channel counts by state, $\boldsymbol{\mu}_0 = \mathbb{E}[\mathbf{N}_0]/N_{\text{ch}}$ and $\boldsymbol{\Sigma}_0 = \text{Cov}(\mathbf{N}_0)/N_{\text{ch}}$, so $\boldsymbol{\mu}_0$ is a probability vector and $\boldsymbol{\Sigma}_0$ is a covariance divided by the channel count. The factors of N_{ch} that appear below are what carry these onto the scale of the recorded current, and they are the reason the rank-one correction of Eq. 13 carries one and the correction to $\boldsymbol{\mu}$ does not. The predicted mean of the interval-averaged current is

$$\bar{y}^{\text{pred}} \;=\; N_{\text{ch}}\,\boldsymbol{\mu}_0^\top \bar{\boldsymbol{\gamma}}_0, \tag{9}$$

and its predicted variance decomposes into three independent contributions,

$$\sigma^2_{\text{pred}} \;=\; \epsilon^2_\Delta \;+\; N_{\text{ch}}\,\widetilde{\boldsymbol{\gamma}^\top \boldsymbol{\Sigma} \boldsymbol{\gamma}} \;+\; N_{\text{ch}}\,\boldsymbol{\mu}_0^\top \boldsymbol{\sigma}^2_{\bar{\gamma}_0}. \tag{10}$$

The first term ϵ^2_Δ is the instrumental noise over the window, which carries the acquisition averaging and decreases with Δ; its parametric form is given in Methods. The second term is the population gating variance, the fluctuation of the window current transmitted through the occupancy covariance. The tilde on it is an operator rather than a decoration, and the symbols under it are not the K-vector $\bar{\boldsymbol{\gamma}}_0$ and the $K \times K$ matrix $\boldsymbol{\Sigma}$ used elsewhere in this section: it denotes a bilinear form taken over boundary *pairs*, contracted back to a scalar. Writing $\boldsymbol{\Sigma}^{\text{bnd}}_{0\to\Delta}$ for the per-channel covariance of the boundary-pair occupancies,

$$\begin{aligned}
\widetilde{\boldsymbol{\gamma}^\top \boldsymbol{\Sigma} \boldsymbol{\gamma}} \;&=\; \sum_{i_0,i_t} \sum_{j_0,j_t} \overline{\Gamma}_{i_0\to i_t} \left(\boldsymbol{\Sigma}^{\text{bnd}}_{0\to\Delta}\right)_{(i_0,i_t),(j_0,j_t)} \overline{\Gamma}_{j_0\to j_t} \\
&=\; \bar{\boldsymbol{\gamma}}_0^\top \big(\boldsymbol{\Sigma}_0 - \text{diag}(\boldsymbol{\mu}_0)\big) \bar{\boldsymbol{\gamma}}_0 \;+\; \boldsymbol{\mu}_0^\top \big(\mathbf{H}\mathbf{1}\big),
\end{aligned} \tag{11}$$

where $H_{i_0 i_t} = \overline{\Gamma}^2_{i_0\to i_t} P_{i_0\to i_t}(\Delta)$ and $\mathbf{1}$ is the vector of ones. The first line is the definition and the second is what is evaluated: only K-vectors and $K \times K$ matrices appear, so the $K^2 \times K^2$ boundary covariance is never formed. The distinction matters for reading Eq. 10, because taking the tilde to mean $\bar{\boldsymbol{\gamma}}_0^\top \boldsymbol{\Sigma}_0 \bar{\boldsymbol{\gamma}}_0$ keeps the first summand and drops the second, which is the single-channel spread of the interval mean across boundary pairs and does not vanish when the occupancy covariance does. The general operator, of which this is the bilinear case, is given in Appendix 1. The third term of Eq. 10 is the within-window per-channel variance, with $\boldsymbol{\sigma}^2_{\bar{\gamma}_0}$ the start-conditioned interval variance built from $\overline{V}_{i_0\to i_t}$.

Writing $\mathbf{g} = \text{Cov}(\text{end-of-window occupancy}, \bar{y})/N_{\text{ch}}$ for the per-channel cross-covariance between the interval current and the occupancy at the end of the window, and $\delta = \bar{y}^{\text{obs}} - \bar{y}^{\text{pred}}$ for the innovation, the Gaussian update of the occupancy is the scalar Kalman correction

$$\boldsymbol{\mu}^{\text{post}} = \boldsymbol{\mu}^{\text{prop}} + \frac{\mathbf{g}}{\sigma^2_{\text{pred}}}\,\delta, \tag{12}$$

$$\boldsymbol{\Sigma}^{\text{post}} = \boldsymbol{\Sigma}^{\text{prop}} - \frac{N_{\text{ch}}\,\mathbf{g}\mathbf{g}^\top}{\sigma^2_{\text{pred}}}. \tag{13}$$

The factor N_{ch} enters the covariance downdate but not the mean correction because $\boldsymbol{\Sigma}$ is carried on the population scale, the same N_{ch} that multiplies the gating variance in Eq. 10, whereas $\boldsymbol{\mu}$ and the gain $\mathbf{g}$ are per-channel objects, so the rank-one correction to $\boldsymbol{\Sigma}$ must be rescaled by N_{ch} to sit on its scale while the correction to $\boldsymbol{\mu}$ needs no such factor. In a recursive member the posterior pair $(\boldsymbol{\mu}^{\text{post}}, \boldsymbol{\Sigma}^{\text{post}})$ becomes the prior for the next window; in a non-recursive member it is not carried forward, and each window's predictive is propagated from the initial condition alone. Reusing the posterior as the next prior is itself a closure: it is exact only while the end marginal is a sufficient summary of the boundary posterior, which the Gaussian projection does not guarantee. For this reason the recursive members' Gaussian information is a model-based quantity and is not claimed to be the exact information of the process; how far it departs from the truth is what the Diagnostics measure.

Two algebraic differences separate the members that sit at $\text{av} = 1$ and $\text{av} = 2$. The first is the form of the third term of Eq. 10. `MR` carries the total per-start-state interval variance, the spread of the interval-averaged conductance of a channel known only to have started in i_0, while `VR` and `IR` carry the residual variance left once the end state is also conditioned on, $\sum_{i_t} P_{i_0\to i_t}(\Delta)\,\overline{V}_{i_0\to i_t}$. By the law of total variance the two forms differ by the spread of the interval mean across end states,

$$\underbrace{\sum_{i_t} P_{i_0\to i_t}(\Delta)\left[\overline{V}_{i_0\to i_t} + \left(\overline{\Gamma}_{i_0\to i_t} - (\bar{\gamma}_0)_{i_0}\right)^2\right]}_{\text{total, MR}} = \underbrace{\sum_{i_t} P_{i_0\to i_t}(\Delta)\,\overline{V}_{i_0\to i_t}}_{\text{residual, VR, IR}} + \mathrm{Var}_{i_t}\left[\overline{\Gamma}_{i_0\to i_t} \mid i_0\right], \tag{14}$$

so `MR` charges to the observation variance a spread that the end state, once conditioned on, would resolve. The second difference is the boundary cross-covariance term

$$N_{\mathrm{ch}}\,\widetilde{\boldsymbol{\gamma}^{\top}\boldsymbol{\Sigma}\boldsymbol{\gamma}} \tag{15}$$

that enters the gain $\mathbf{g}$ of Eqs. 12 and 13, which `IR` retains and both $\mathrm{av} = 1$ members drop. Stepping `MR` to `VR` changes the variance form and nothing else, and stepping `VR` to `IR` restores the boundary term in the gain and changes nothing else, so the two halves of the `R` to `IR` step are separated and can be measured one at a time. Their consequences for the predicted variance across the regime grid are characterized in the Results.

The top of the ladder

The exact likelihood conditions on the full trajectory inside each window and is intractable, which is why the two closures above are needed at all. The same stochastic model that makes the exact likelihood intractable can be simulated forward exactly, one trajectory at a time, so the simulation realizes the object at the top of the ladder and serves as the ground truth against which every rung below it is judged.

Diagnostics: testing a likelihood against its simulation

An approximate likelihood is a misspecified model, and misspecification leaves a classical signature. At the true parameters the score no longer has zero mean, so the estimator the likelihood defines is biased; and the covariance of the score no longer equals the negative expected curvature, so the uncertainty the likelihood reports is wrong. These are the two Bartlett identities, and their failure is the content of the information-matrix-equality diagnostic of ***White*** (***1982***) and of the sandwich covariance of ***Huber*** (***1967***); ***Godambe*** (***1960***). In the classical setting those identities can only be tested, through a statistic with a known finite-sample defect, because the data-generating law is unknown. Our setting removes that limitation. The forward process is an exact continuous-time Markov chain (CTMC) and we simulate it exactly, so we hold unlimited data drawn from the very law the likelihood claims to describe, at parameters we set. The classical test therefore becomes a direct measurement: the anchor curvature is computed analytically and the score covariance is estimated by Monte Carlo over independent simulated recordings, with no null hypothesis to reject. The apparatus is standard; what is new here is applying it in this domain and reading its output as a measured property of each approximation rather than as a hypothesis to accept or reject.

We read three quantities off the simulation, in increasing strength, and each is read at a parameter vector that has to be named because there are two of them. The first is the simulation truth θ_{sim}, the point the recordings were generated from, which is known here and never known in an experiment. The second is the pooled optimum θ_{pool}, the maximum-likelihood fit of the member under test to all the recordings of one cell at once; it is not known in closed form and is located numerically, by Gauss–Newton with Levenberg damping warm-started at the truth, with a near-zero gradient certifying that the point reached is stationary (Methods). Were the likelihood correctly specified the two would agree up to sampling error. They do not, and their difference $\theta_{\mathrm{pool}} - \theta_{\mathrm{sim}}$ is the misspecification bias itself. Everything below is computed at both anchors and every figure states which one it shows, because they answer different questions: at θ_{sim} a nonzero mean score is visible at all, whereas at θ_{pool} the score vanishes by construction, so the information comparison is not contaminated by a displaced gradient.

The first is the standardized residual. With μ_t and σ_t^2 the likelihood's own predictive mean and variance for interval t, the residual $r_t = (y_t - \mu_t)/\sigma_t$ must have mean zero, unit variance, and no autocorrelation across intervals if the one-step predictive distribution is correct *at the parameters it is evaluated at* (Figure 3 draws the second and the third; the first is carried in a supplement, where it separates no member and so returns no verdict). This probes the predictive one interval at a time. It is the cheapest of the three and the only one an experimentalist can run on a real recording, because it needs no ensemble and no known truth; but it does need a fitted point, since at an arbitrary parameter vector the three properties fail for reasons that have nothing to do with whether the approximation is any good. Running it at the fit has a cost of its own. For a Gaussian likelihood the score is a weighted combination of exactly the quantities r_t and $r_t^2 - 1$, so setting it to zero imposes as many linear constraints on the residual moments as there are free parameters, and a fit that leaves the baseline and the noise level free absorbs most of the first two properties into itself. What no fit absorbs is the temporal structure. Residual whiteness is therefore the part of this check that stays informative at the optimum, and it is not optional: residual autocorrelation is the same signal that the strongest check quantifies below, though it is a conservative reading of it, running 1.6 to 1.7 times smaller than the score autocorrelation for the members in which either is large.

The second is the score. Writing $\ell(\theta)$ for the total log-likelihood and $S(\theta) = \nabla_\theta \ell(\theta)$ for its gradient, correct specification requires $\mathbb{E}[S(\theta_{\text{sim}})] = 0$ at the simulation truth. A nonzero mean score is what biases the estimate, and projected through the parameter covariance it becomes a displacement in parameter units, $b = \mathbf{H}^{-1}\,\mathbb{E}[S]$, the reader-facing statement of how far the estimate is pushed (Figure 3). This projection is a first-order approximation, from the expansion of the score at the optimum given below. Unlike the residual, this one cannot be run on an experiment at all, and for two reasons rather than one. It needs a known truth, and it needs an ensemble to take the expectation over. At the only point an experimenter has, their own fit, the score vanishes by construction and the check is empty. The simulator supplies both, and the confidence interval narrows as the number of simulated recordings grows.

The third and strongest is the information itself. Correct specification requires $\mathrm{Cov}[S] = \mathbf{H}$, the information the likelihood reports, the covariance being taken over independent recordings of the same cell. That is the second thing an experiment cannot supply: one recording gives one score, and a covariance cannot be estimated from a single draw. An approximation can pass the residual and score checks and still violate this one by an order of magnitude. The two failures are independent: whether the estimate is biased is governed by how the conductance is modelled over the interval, while whether the reported information agrees with the information delivered is governed by the recursion. We turn all three checks on the whole cost ladder set out in Theory, from least squares on the deterministic mean current (LSE) at the bottom to the boundary-conditioned filter at the top. The range they have to cover is wide. The members that carry no recursion misreport the information grossly, and among the recursive members the interval conditioning leaves distortions small enough that a diagnostic has to be sharp to resolve them, so the same instrument has to remain informative across both. Because the failure is directional, we record it as a matrix,

$$\mathbf{C} = \mathbf{H}^{-1/2}\,\mathbf{J}\,\mathbf{H}^{-1/2},$$

with $\mathbf{J} = \mathrm{Cov}(S)$ the covariance of the total score, cross-interval terms included. This definition is exact algebra given $\mathbf{H}$ and $\mathbf{J}$, and under correct specification $\mathbf{J} = \mathbf{H}$ so $\mathbf{C} = \mathbf{I}$ exactly. We fix the direction by convention: $C_{ii} > 1$ means the likelihood under-reports the uncertainty in parameter i and is over-confident, $C_{ii} < 1$ means it over-reports and is conservative. The eigen-directions of $\mathbf{C}$ say which combinations of parameters are misreported, which a scalar summary cannot recover.

The anchor $\mathbf{H}$ is the likelihood's own Gaussian Fisher information,

$$\mathbf{H} = \sum_t \mathbb{E}\left[\frac{(\nabla\mu_t)(\nabla\mu_t)^\top}{\sigma_t^2} + \frac{(\nabla\sigma_t^2)(\nabla\sigma_t^2)^\top}{2\sigma_t^4}\right], \tag{16}$$

formed from the same predictive moments and their parameter derivatives that the score already requires, at no extra cost. For a macroscopic algorithm the likelihood is Gaussian by construction, so this is the model's own Fisher and $\mathbf{C}$ is the information-matrix-equality test carried out in the model's own frame. Two properties follow. The Gaussian Fisher is positive semidefinite by construction, so the diagnostic never has to be rescued from a numerically indefinite anchor. And it is analytic, so it does not inherit the finite-difference noise of a numerical Hessian, which in this regime is severe: the finite-difference Fisher is widely indefinite per replicate, and only pooling across replicates recovers a definite matrix. By design the anchor is the Gaussian Fisher; the numerical Fisher we compute alongside it serves to gauge how faithful that Gaussian anchor is. Comparing the approximation to its own Fisher is not circular. $\mathbf{H}$ is the information the likelihood claims, $\mathbf{J}$ is the information it delivers against data from the true process, the two are computed from different objects, and their disagreement is the definition of misspecification.

The bottom rung of the ladder needs a qualification at the point where the diagnostic is defined, because its predictive variance is not a gating quantity. Least squares fits the deterministic mean current under a single variance held constant over the whole recording, so the information it reports, $\mathbf{H}$ as written above with σ_t^2 replaced by one constant σ^2, is the information of a homoscedastic model. The exact process violates that assumption: the variance of a recorded interval rises and falls with the open population over the course of the concentration jump, and it also depends on the channel number, so the constant-variance premise is wrong by construction in every cell we simulate. For LSE the quantity $\mathrm{Cov}(S)$ read against the reported information therefore measures the size of that break. It is not an information-matrix-equality failure of the same kind as a gating-aware likelihood's, where the predictive variance does follow the gating and the discrepancy is attributable to how the gating is approximated within the interval and across intervals. The Results quote LSE's ratio with that reading, and they do not read an LSE ratio and a gating-aware ratio as two values of one comparable quantity.

The same matrix repairs what it measures. The sandwich covariance

$$\Sigma = \mathbf{H}^{-1}\,\mathbf{J}\,\mathbf{H}^{-1} = \mathbf{H}^{-1/2}\,\mathbf{C}\,\mathbf{H}^{-1/2}$$

is the corrected parameter covariance: it substitutes the information the likelihood actually carries for the information it reported. This step, and the bias vector b above, are first-order approximations, from a Taylor expansion of the score at the optimum with the empirical Hessian replaced by its expectation $-\mathbf{H}$ (see Supplementary Information). We verify the correction directly, without leaning on the expansion, by comparing the sandwich-corrected ellipse against the empirical covariance of the maximum-likelihood estimates (MLEs) over the simulated recordings (Figure 2). A diagnostic that also corrects the error bars it flags is a stronger result than one that only scores algorithms.

The distortion splits into two physically distinct parts (Figure 4–figure supplement 1). The sample distortion $\mathbf{C}_s = \mathbf{H}^{-1/2}\,\mathbf{J}_s\,\mathbf{H}^{-1/2}$, built from the within-interval score covariance $\mathbf{J}_s = \sum_t \mathrm{Cov}(s_t)$ with $s_t = \nabla_\theta \ell_t$ the score of interval t, measures per-sample departure from Gaussianity; to the leading corrections it is the third and fourth cumulants of the interval current, the non-Gaussian shape the Gaussian predictive cannot carry. The correlation distortion $\mathbf{R} = \mathbf{J}_s^{-1/2}\,\mathbf{J}\,\mathbf{J}_s^{-1/2}$ measures what the within-interval covariance misses, the cross-interval correlation of the score, which is exactly the local time correlation of the current that ***Milescu et al.*** (***2005***) named as the dominant error of non-recursive fitting; $\mathbf{R} = \mathbf{I}$ when the score carries no cross-interval dependence. The figures resolve the same pair in time and one parameter at a time, and the notation follows the matrices. For the parameter under discussion, F_t is the Gaussian Fisher information of interval t alone, so that the corresponding diagonal entry of $\mathbf{H}$ is $\sum_t F_t$; $J_t = \mathrm{Var}(s_t)$ is the variance of that interval's score. Accumulated to interval T they are $F_T = \sum_{t\le T} F_t$ and $J_T = \mathrm{Var}\big(\sum_{t\le T} s_t\big)$, and it is the second of these, not the sum of the J_t, that the reported information has to match. The two statements $J_t = F_t$ and $J_T = F_T$ are therefore not equivalent, and their divergence is the whole of what Figure 3 measures. These compose exactly. The exact identity is

$$\mathbf{C} = \mathbf{K}\,\mathbf{R}\,\mathbf{K}^{\top}, \qquad \mathbf{K} = \mathbf{H}^{-1/2}\,\mathbf{J}_s^{1/2},$$

which holds as algebra for any $\mathbf{H}$ and $\mathbf{J}_s$, since substituting the definitions collapses the chain. The factor $\mathbf{K}$ satisfies $\mathbf{K}\mathbf{K}^{\top} = \mathbf{C}_s$, so it is a legitimate square-root factor of the sample distortion, but it is not the symmetric principal square root, and the form $\mathbf{C} = \mathbf{C}_s^{1/2}\,\mathbf{R}\,\mathbf{C}_s^{1/2}$ holds only where $\mathbf{H}$ and $\mathbf{J}_s$ commute. What composes with no assumption at all is the information volume,

$$\log\det\mathbf{C} = \log\det\mathbf{C}_s + \log\det\mathbf{R},$$

an exact identity because the determinant is multiplicative. The per-parameter diagonal, the trace, the eigenvalues and the Frobenius norm do not compose across the split; only the volume does, and only the volume is decomposed in the maps.

The correlation distortion has a reader-facing translation. For any additive statistic X_t with sum $L = \sum_t X_t$, the ratio $\kappa = \mathrm{Var}(L)\big/\sum_t \mathrm{Var}(X_t)$ is unity when the terms are independent and larger when they are positively correlated, and it converts to an effective sample size $T_{\mathrm{eff}} = T/\kappa$: a non-recursive likelihood believes it has T independent intervals when it effectively has T/κ. This is the number from the diagnostic most worth quoting, and it is exact by definition. When $\mathbf{R} \approx \kappa\mathbf{I}$ the temporal error is isotropic and this scalar captures it; when $\mathbf{R}$ is anisotropic, or when $\mathbf{C}_s \ne \mathbf{I}$, the distortion cannot be reduced to a sample-size correction and the full matrix is needed.

A matrix is not a number, and two scalars summarise it without pretending to replace it. Let $\lambda_1, \dots, \lambda_d$ be the eigenvalues of $\mathbf{C}$ on the retained subspace. Their logarithms have a mean $\bar{e} = d^{-1}\sum_i \log\lambda_i$ and a variance $s^2 = \mathrm{Var}_i(\log\lambda_i)$, and the two quantities reported throughout are their exponentials,

$$m = e^{\bar{e}} = \Big(\prod_i \lambda_i\Big)^{1/d}, \qquad a = e^{s}, \tag{17}$$

so that m is the typical factor by which the reported information is wrong, the geometric mean of the per-direction factors, and a is the multiplicative spread of those factors from direction to direction. A member is drawn as a point at m with a bar from m/a to $m\,a$, which is why the axes are logarithmic and why the spread is written as a factor rather than as a plus-or-minus. Both are one at correct specification. The single scalar that combines them is the affine-invariant Riemannian distance from the identity, $D_{\mathrm{AI}} = \big(\sum_i \log^2\lambda_i\big)^{1/2} = \sqrt{d\,(\bar{e}^2 + s^2)}$, which is used where a single number is wanted and is reported nowhere in place of the pair. In error-bar units take the square root of any of these, since the distortion is a ratio of informations: a distortion of 1.15 is a 7% error on a confidence interval, 2 is 41%, and 4 turns a nominal 95% interval into about 68%.

Two conventions govern the edge cases. The direction is fixed as above, $C_{ii} > 1$ being over-confidence and $C_{ii} < 1$ conservatism, stated once so every verdict reads the same way. And the anchor can be near-singular where a parameter is unidentifiable, which by design happens once the open population relaxes (Figure 3–figure supplement 1); there we diagonalize $\mathbf{H}$ and retain the eigenmodes whose eigenvalue exceeds a tolerance set relative to the largest, $\lambda_i > \lambda_{\max}\cdot 10^{-10}$, evaluate every inverse power on that retained subspace alone, and leave the quantity undefined, not finite, when the retained subspace is empty. The dimension d of Eq. 17 is the dimension of that subspace, so a cell where a direction has been dropped is summarised over fewer directions than one where none has. Undefined is not zero, and the maps render those cells in grey to keep the two apart.

Results

The figures are ordered so that each one answers the objection the one before it raises. Figure 1 shows that the members of the cost ladder do different arithmetic on the same recording, which invites the question of whether that arithmetic changes what an experimenter takes away. Figure 2 answers it at one design cell, where the members recover the parameters and report intervals that differ by more than an order of magnitude. That raises the question of whether the fault lies in the likelihood or in the fit, so Figure 3 puts the question to the likelihood itself, scoring one common set of recordings interval by interval, and locates the failure in how the information accumulates over time. One cell is not a result, so Figure 4 repeats both moments over the plane of channel number against instrumental noise, with the acquisition interval resolved inside each cell. A member whose reported interval is honest over most of that plane invites the question of where it stops being so, and Figure 5 answers it on that corner, decomposing the departure into the two discarded moments that cause it. Figure 6 collects the answers into a map of what a macroscopic recording can be asked for.

Throughout, N_{ch} is the number of channels in the patch, k_{on} and k_{off} the opening and closing rates, i the unitary current, Δ the acquisition interval and $\tau = 1/k_{\mathrm{off}}$ the closing time constant, so that $\Delta \cdot k_{\mathrm{off}}$ is the interval in units of τ. The instrumental noise is reported as the label the runs and the figure axes carry; the dimensionless noise $\tilde{S}$, the instrumental noise power accumulated over one channel time constant measured against the unitary current, is one tenth of that label (Methods).

One filter step along the cost ladder

Figure 1 puts one turn of the filter cycle for the four body members side by side, on a single simulated recording of a patch of $N_{\mathrm{ch}} = 20$ channels. It fixes the vocabulary and carries no measured claim.

Least squares (LSE) fits the deterministic mean current and assigns one constant variance $\hat{\sigma}^2$ to every residual, estimated as the residual sum of squares over the record. Its predicted band therefore has the same width at every interval, whether the channels are gating or not, which is the homoscedastic premise of the classical fit made visible on the trace. The non-recursive member (NR) replaces that single constant with the per-interval gating variance the model predicts, so its band follows the open population, and its update row reads as an open loop: the occupancy carried into each interval comes from the initial condition and never from the data. The recursive member (R) adds the Bayesian update, on a conductance taken at a single instant. The boundary-conditioned member (IR) conditions the interval-averaged conductance on the channel states at both ends of the interval, which lets the covariance between them enter the gain. Reading left to right, each column adds one operation to the column before it, and the cost of a likelihood evaluation rises with it.

Recovery at one design cell

Figure 2 turns the ladder into a statement about reported uncertainty at one cell: $N_{\mathrm{ch}} = 100$, noise label 0.1 ($\tilde{S} = 0.01$), $\Delta \cdot k_{\mathrm{off}} = 0.1$. Ten thousand independent recordings are partitioned into groups of ten, each group is fitted by maximum likelihood under each member, and the scatter of the resulting thousand estimates is compared against the covariance that member reports from its own Gaussian Fisher information.

The two objects disagree by an order of magnitude at the foot of the ladder and by nothing at its top, but the fall between them is not monotone. Taking the ratio of the area of the empirical 95% ellipse to the area of the reported one, and averaging geometrically over the six pairs the four informative parameters form, the open-loop member reads 11.41 (per-pair range 10.04 to 14.83), the recursive instantaneous member 1.17 (1.08 to 1.32), and the boundary-conditioned member 1.01 (0.98 to 1.05). On the joint four-parameter ellipse the same three read 154.65, 1.56 and 1.12. Per parameter, the diagonal of the distortion matrix $\mathbf{C}$ at this cell runs $13.24/17.32/10.04/9.91$ for NR, $1.39/1.51/1.16/1.16$ for R and $1.05/1.13/1.00/0.90$ for IR, on k_{on}, k_{off}, i and N_{ch} in that order. A likelihood that carries no recursion therefore under-reports the width of the parameter distribution by a factor of ten to fifteen, while the recursive members are wrong by tens of percent. The one-

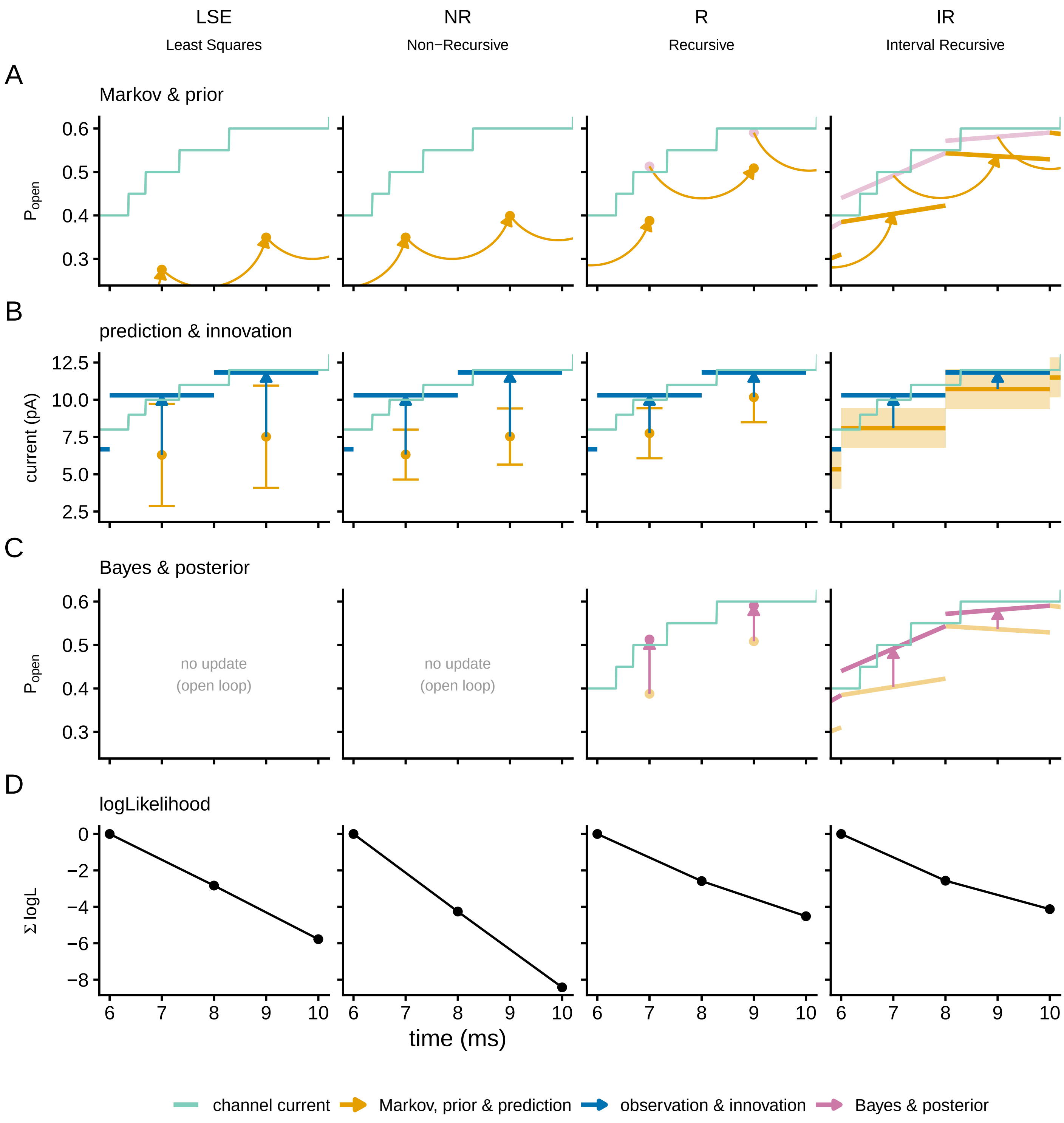

Figure 1. One filter step along the cost ladder. The current from a patch of $N_{\mathrm{ch}} = 20$ two-state channels, recorded as an interval average with white instrumental noise, is scored by each of the four body members (columns): least squares on the deterministic mean current (`LSE`), the open-loop gating likelihood (`NR`), the recursive filter on instantaneous samples (`R`), and the boundary-conditioned filter (`IR`). Rows follow one turn of the cycle: (A) the prior open probability from the Markov step, (B) the predicted current with its spread and the innovation, (C) the Bayesian posterior, which for the two non-recursive columns reads as no update, and (D) the per-step log-likelihood accumulated over the window and reset at the window start. Two measurement windows are shown (6 to 10 ms of a 12 ms recording). `LSE` is drawn in `NR`'s grammar, a point mean with an error bar, so that its constant least-squares variance can be read against `NR`'s per-interval gating variance. Colour encodes the filter phase and is shared across rows: green the simulated channel current, orange the predict phase, blue the observation and the innovation, purple the update, black the log-likelihood. A quantity carried in from the neighbouring interval is drawn hollow or dashed. Three of the seven rungs are not drawn, because the form of the cycle does not change for them; the one step that does change form, the gain acquiring the boundary term, is the `R`-to-`IR` step drawn here. Appendix 1 writes all

endpoint members do not sit between the two recursive ones, and this is where the ladder stops being an ordering. On the kinetic pair the recursive instantaneous filter reads 1.32 while `MR` reads 1.97 and `VR` reads 2.19, and only then does `IR` return to 1.02; the amplitude pair runs 1.09, 1.53, 1.77, 1.00 and the noise pair 1.06, 1.29, 1.43, 0.99 in the same order. Conditioning the conductance on one endpoint therefore costs calibration rather than buying it, in every pair and in both directions of the variance switch, and what recovers it is the second endpoint entering the gain (Appendix 1). The two intermediate rungs are kept in the figure for exactly this reason: they are the step from `R` to `IR` taken in its two algebraic halves, and the half that changes only the interval variance moves the answer the wrong way.

What the acquisition average buys on its own is separable here, because one member isolates it: the one-endpoint non-recursive member conditions the conductance on the interval instead of on an instant and adds nothing else, no update and no second endpoint. Against the open-loop member it improves every pair, from 14.83 to 12.59 on the kinetic pair and from 10.04 to 9.00 on the amplitude pair, and on the noise pair it improves by nearly a factor of three, from 8.84 to 3.28. The reason is in the first moment rather than the second. The open-loop member carries no interval-mean conductance variance, so the gating variance it predicts is too small, and the fit makes up the difference by inflating the one free parameter that can absorb it: its noise estimate lands a factor of 2.2 above the truth. Restoring that term removes the need to compensate, and the noise estimate returns to within 0.005 in $\log_{10}$. Modelling the acquisition average therefore buys the noise level, and it buys it with no recursion at all.

The distortion is measurable and it is also removable. Replacing each reported covariance by the sandwich, which substitutes the observed covariance of the score for the information the likelihood claimed, returns every gating-aware member of the ladder to within about a tenth of one: 1.03 for `NR`, 0.95 for `R`, 1.05 for `IR`, with every parameter pair falling between 0.87 and 1.07, and the two one-endpoint members inside the same band. The correction repairs the members that model the fluctuations, whatever they get wrong about them, and it is least squares alone that it leaves short. The reader-facing form of that repair is coverage rather than area. Reducing the full multivariate distribution of the estimates to the squared Mahalanobis distance from the cloud mean, which is distributed as χ^2 with as many degrees of freedom as free parameters if the reported covariance is right, the nominal 95% region of `IR` covers 0.914 of the estimates under its own Fisher covariance and 0.947 to 0.949 under the sandwich, measured at $N_{\mathrm{ch}} = 10$ and noise label 0.05 over a thousand fits in six parameters at group size 100 (Figure 4–figure supplement 2). That one correction repairs six of the seven rungs, without being told which of them is wrong or how, is what makes the distortion a usable quantity rather than a complaint.

The noise pair separates the members in a way the other two rows do not, and it does so on the first moment rather than the second. The open-loop member places its instrumental-noise estimate at $\log_{10}\sigma = -3.67$ against a simulated -4, a factor of 2.2 too much noise, which is what a likelihood does when it must absorb gating variance it does not model into the one variance it has. Every other member sits within 0.04 of the truth, the one-endpoint non-recursive member included, so this is not a cost of recursion or of the interval average but of modelling the fluctuations at all.

Least squares carries a column in the kinetic pair and none in the other two, and the reason is identifiability rather than a gap in the runs. Its grid configuration holds the unitary current and the instrumental noise level fixed at the values the recordings were simulated from (Methods), so in the amplitude pair its estimates would lie exactly on the line $i = 1$ and in the noise pair exactly on $\sigma = 10^{-4}$: the empirical and the reported ellipse would both have zero area and the ratio the panel reports would be $0/0$. The mean current depends on the channel number and the unitary current only through their product, and it is the gating fluctuations, whose variance carries a second power of the unitary current, that separate the two. A method that models no fluctuations estimates the product, and fixing the unitary current names that product after one of its factors. Where least squares does compete, on the pair the deterministic transient determines, it under-reports by 14.8, which is the open-loop member's own figure, and the sandwich does not repair it: corrected, it still reads 1.63, while the other six members corrected fall between 0.96 and 1.07 and pair.

The calibration cascade in time

The miscalibration of Figure 2 is a property of the likelihood and not of the optimiser, and Figure 3 shows where in the recording it is produced. One common set of a thousand recordings at the same cell ($N_{ch} = 100$, noise label 0.1, $\Delta{\cdot}k_{off} = 0.1$) is scored interval by interval by each member, with no fitting involved. The least-squares arm here is the six-parameter configuration, which keeps the unitary current and the noise level free so that its per-interval quantities align index for index with the macroscopic dumps (Methods); nothing measured in this subsection may be carried across to the four-parameter configuration of the later figures.

The members separate first on the log-likelihood they assign to the same data. The ensemble-mean total is -262 ± 0.73 for `LSE`, -230 ± 0.83 for `NR`, -147 ± 0.20 for `R` and -143 ± 0.24 for `IR`, a spread of 118.9 nats from the bottom of the ladder to the top. The ordering follows the ladder, and most of the span is bought by modelling the gating fluctuations at all.

The hinge of the argument is that per interval every member reports its information correctly. The variance of the per-interval score matches the per-interval Fisher information for all four, and the median of $\log_{10}(J_t/F_t)$ over the informative steps is $+0.184$ for `LSE`, -0.038 for `NR`, -0.181 for `R` and $+0.006$ for `IR` on k_{on}, with -0.145 and $+0.033$ for `R` and `IR` on k_{off}. Accumulated over the recording the same ratio reads 14.96 (95% interval 13.86 to 15.93) for `LSE`, 13.63 (12.52 to 14.95) for `NR`, 1.07 (0.975 to 1.17) for `R` and 1.08 (0.995 to 1.17) for `IR`. The two non-recursive members are within a few percent of the identity at each interval and off by more than an order of magnitude once the intervals are summed. The open-loop member also changes the sign of the discrepancy between the per-step and the accumulated statistic, and a per-sample modelling error cannot do that. Least squares does not change sign, sitting above the calibrated value at both levels, which is the signature of its constant-variance premise rather than of accumulation.

The least-squares ratio needs the qualification the Diagnostics section attaches to it. The Gaussian Fisher information of a least-squares fit presumes one constant residual variance, and the exact simulator violates that premise by construction, because the variance of a recorded interval rises and falls with the open population. The factor of fifteen is the size of that break, and it is not an information-matrix-equality failure of the same kind as the open-loop gating likelihood's, where the predictive variance does follow the gating. The two ratios are reported side by side and are not read as two values of one comparable quantity.

The same checks run for the parameters the figure has no colours for, and for the first moment of the residual, in Figure 3–figure supplement 2. Two things there are worth carrying back: the instrumental noise level is the only parameter measured before the agonist arrives, and its score is white even in the members whose kinetic directions are not, so the memory that carries the accumulated error is a property of the gating directions; and least squares has no score in that direction at all, because its noise level is a plug-in from the same record rather than a free parameter.

The mechanism is visible in the same figure. At lag one the autocorrelation of the per-interval score on k_{off} is -0.0043 ± 0.0043 for `IR`, indistinguishable from zero, against 0.191 ± 0.004 for `R`, 0.864 ± 0.0019 for `NR` and 0.856 ± 0.0022 for `LSE`. A positively correlated score makes the variance of its sum exceed the sum of the per-interval informations, which is exactly how the per-interval identity can hold while the accumulated one fails. The temporal correlation the two non-recursive members leave in their score is the local time correlation of the current that ***Milescu et al.*** (***2005***) named as the dominant error of independent-interval fitting, measured here at the level of the inference rather than the data.

Row C of Figure 3 carries a second result. For the members that condition on the past, the information about the channel number and the opening rate reaches the numerical floor within six intervals of the agonist being removed, at 1.4×10^{-21} and 7.9×10^{-44} respectively for `IR` over the last ten intervals: having conditioned on everything up to that point, the filter gains nothing further from watching the channels decay. Two parameters survive the washout, and they survive

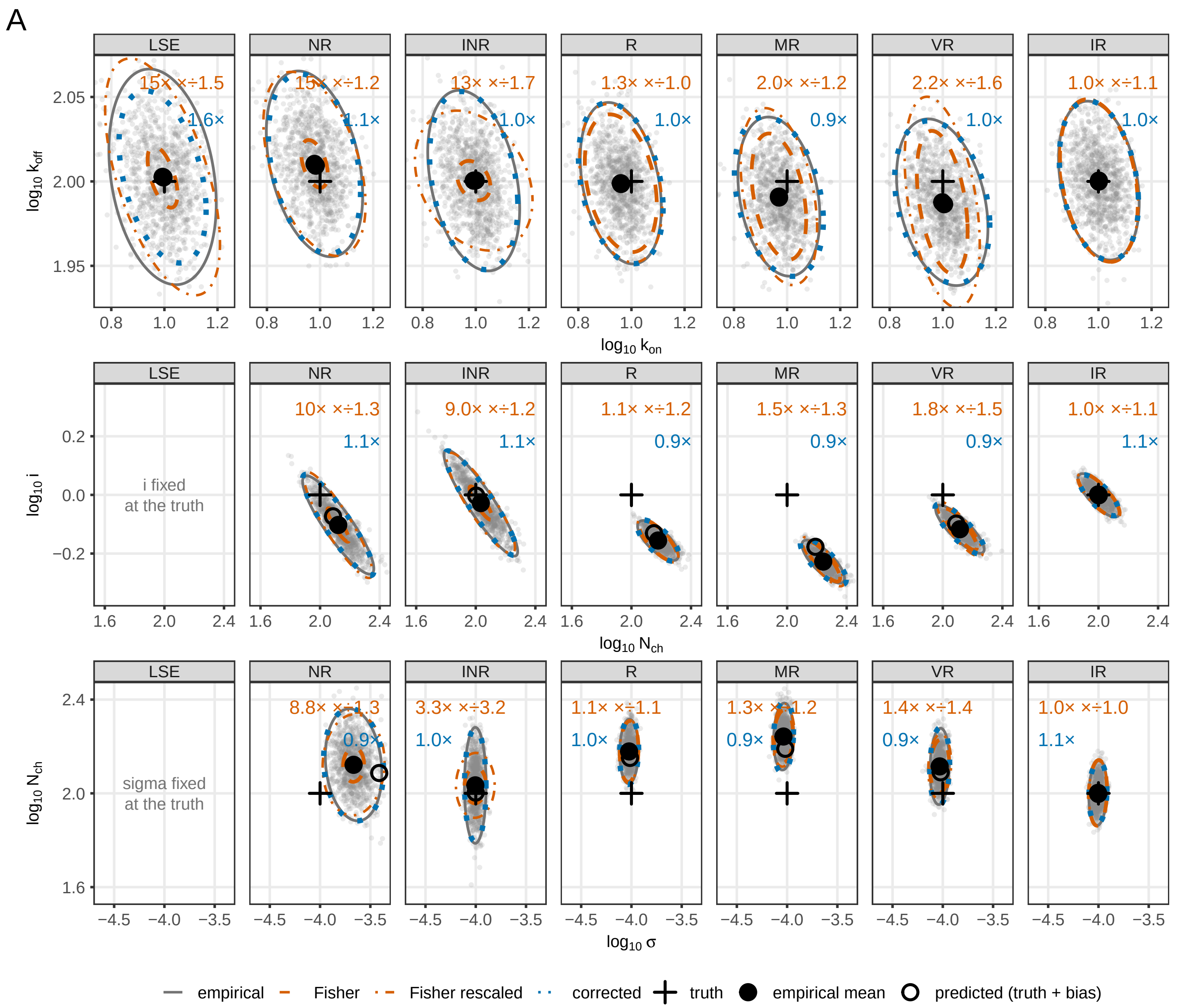

Figure 2. Parameter recovery and the calibration of the reported interval, for the whole ladder at one design cell. $N_{\mathrm{ch}} = 100$, noise label 0.1 ($\tilde{S} = 0.01$), $\Delta \cdot k_{\mathrm{off}} = 0.1$; $10{,}000$ recordings in groups of ten, so one thousand maximum-likelihood estimates per member. Columns are the seven rungs in ladder order, from least squares to the boundary-conditioned filter. Rows are three parameter pairs, each chosen because a different property separates the members in it: the kinetic pair ($\log_{10} k_{\mathrm{on}}$, $\log_{10} k_{\mathrm{off}}$), which the deterministic transient determines and on which every member including least squares competes; the amplitude pair ($\log_{10} N_{\mathrm{ch}}$, $\log_{10} i$), which only the gating fluctuations separate; and the noise pair ($\log_{10} \sigma$, $\log_{10} N_{\mathrm{ch}}$), where σ is the instrumental noise level, a free parameter of every likelihood member, with a simulated value of 10^{-4}. Each panel carries the cloud of estimates and three 95% ellipses drawn at the cloud mean: the empirical covariance, the Fisher covariance the member reports, and the sandwich covariance. The two annotated numbers are the area of the empirical ellipse over the area of the reported one (orange) and over the corrected one (blue), so a member is calibrated when both read 1.0. Markers are the simulation truth, the mean of the cloud, and the truth displaced by the predicted bias. The two least-squares cells that are empty are empty by construction and say so: that configuration holds the unitary current and the noise level fixed at their simulated values, so its estimates would lie on a line and the area ratio would be $0/0$. Covariances are the Gaussian-Fisher family at the pooled estimate, scaled by the reciprocal of the group size. What the third row separates is in the text.

by different routes. The closing rate keeps 1.13, and 81% of what it keeps comes from the mean, which is the shape of the decay its own kinetics govern. The unitary current keeps 0.66, and 71% of what it keeps comes from the variance, which is the gating fluctuation of the channels still open, so it falls by a factor of 36 and then levels at the same order as the closing rate rather than reaching the floor. This is a property of conditioning and not of the macroscopic observable. The two non-recursive members retain 15 to 16% of their pulse information about the channel number through the washout, against 0.5 to 0.8% for the two recursive ones, which is what the mechanism above predicts, since gaining nothing further from the decay requires having conditioned on it.

The design plane, in both moments

One cell is not a result, and the plane carries one result the cell cannot: instrumental noise equalises the members. Where it grows past the gating variance every rung reports honestly, so the differences between them are a property of the regime in which the fluctuations still carry information, and not of the algorithms in general. Everything below is read against that. Figure 4 repeats the two moments over the whole grid, with the members across the columns, the channel count down the rows, and the acquisition interval against the instrumental noise inside each panel. It splits the display by moment rather than by member, so the least-squares arm stays alongside the likelihoods it is being compared with. Half (A) is evaluated at the simulation truth, where a bias is visible; half (B) at the pooled optimum, where the score vanishes by construction and the information comparison is clean (Methods). Both halves are drawn on one scale, the factor by which the report is wrong, so the paleness of (A) against the saturation of (B) is a comparison and not an impression. The least-squares arm here is the four-parameter configuration, which holds the unitary current and the instrumental noise level fixed at the values the recordings were simulated from (Methods), so it holds one column per half, the closing rate only: with the unitary current assumed, the amplitude ridge that limits the other three members does not exist for it, and a channel-number panel would read as least squares winning a contest it did not enter.

More channels do not repair the recursive instantaneous filter. Taking the median over the seven intervals at noise label 0.1, the k_{off} distortion of `R` sits on a flat floor of 1.372, 1.440, 1.439 and 1.42 across N_{ch} 10, 100, 1000 and 10^4, while `IR` converges toward one over the same range, 1.324, 1.099, 1.000, 1.00. The open-loop member moves the other way, 21.92, 16.83, 31.32 and 76.52, and the least-squares arm holds a floor an order of magnitude above one, 11.79, 13.06, 13.04 and 12.97. In the channel-number direction the two recursive members separate in sign as well as in size: `IR` runs 0.851, 0.936, 0.987, 1.00 and stays on the conservative side, while `R` crosses from conservative to over-confident, 0.866, 1.100, 1.230, 1.26, and keeps rising. The two members agree at the fewest channels and separate from there, so the 10^4 column is the informative one.

The first moment carries a separate cost. The recursive instantaneous filter has a bias in the channel number of 0.126, 0.148, 0.138 and 0.138 in $\log_{10}$ units across the same four decades, close to a 35 to 40% error that does not shrink as channels are added; the open-loop member carries 0.073 to 0.084 in the same direction, and the boundary-conditioned member is at or near zero everywhere. Least squares has a rate bias of essentially zero over the same plane. That concession is worth making plainly: the classical fit recovers the rates, and what it gets wrong is how well it says it recovered them.

Read together, the two moments say that the two axes of the family act on different failures and that neither substitutes for the other. Averaging over the acquisition interval is what removes the amplitude bias: the median absolute bias in the channel number, in $\log_{10}$ units, is 0.099 for the open-loop member and 0.110 for the recursive instantaneous one, against 0.003 and 0.002 for their interval-averaged counterparts, so recursion alone does not remove it and slightly worsens it. Recursion is what removes the distortion: the same step drops the closing-rate distortion from tens to about 1.4, and it reaches one when the recursion conditions on both ends of the interval. The same factorial appears a second time, in a different measurement and at a single cell, in the four corner values of Figure 3: interval averaging alone leaves the per-interval residual calibrated

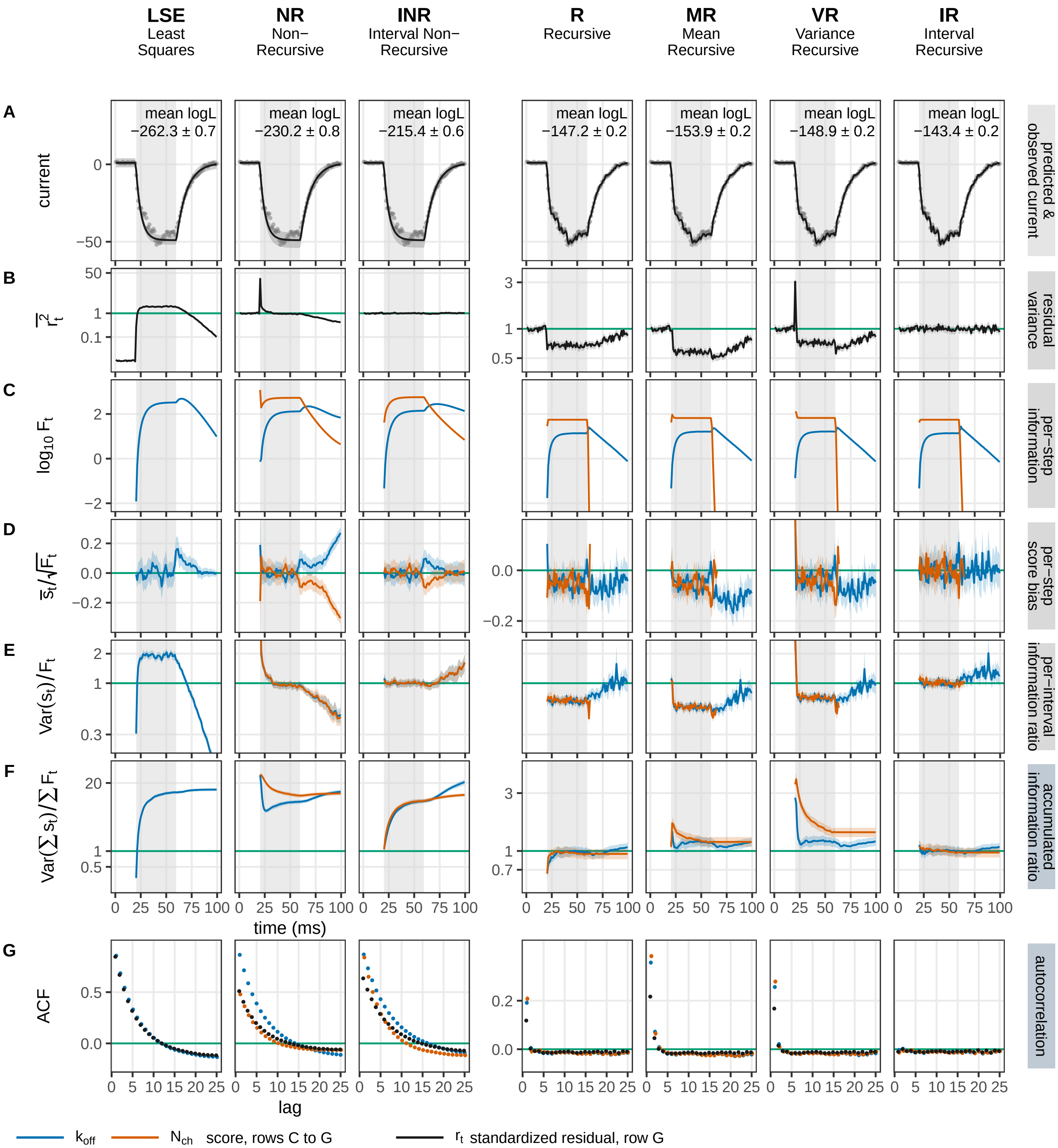

Figure 3. The calibration cascade in time, over the whole implemented family. One set of $1{,}000$ recordings ($N_{\mathrm{ch}} = 100$, noise label 0.1, $\Delta \cdot k_{\mathrm{off}} = 0.1$) is scored by each of the seven members (columns, in cost-ladder order), with no fitting. Rows A and B are properties of the data, in grey; rows C to G are per parameter, the closing rate and the channel number in colour. (A) An example recording with the predicted current, its band the filter's own predictive ± 1 s.d., annotated with the ensemble-mean log-likelihood and its standard error. (B) The mean squared standardized residual, one when the predictive variance is right. (C) The per-interval Fisher information, so where each parameter is measured. (D) The mean per-interval score over $\sqrt{F_t}$, zero when unbiased and standardized so parameters with different units share an axis. (E) The per-interval information ratio $\mathrm{Var}(s_t)/F_t$, one at each interval. (F) The same ratio accumulated, J_T/F_T. (G) The autocorrelation against lag of the standardized residual, in black, and of the score for each parameter, in its colour; the score's is why (E) holds where (F) does not, and the residual's is the same memory in the data and the only quantity here an experimentalist can compute on a real recording. Read across the columns the figure is a factorial on the two switches of the lattice, and the four cells of that factorial are the values to compare: interval averaging alone (`INR`) reads 0.9995 on (B) against 20.8 on (F), recursion alone (`R`) reads 0.782

at 0.9995 and the accumulated information ratio at 20.8, recursion alone inverts that pair to 0.782 and 1.07, and the member with both is at one on each. Two quantities measured in two ways, one resolved in time at one cell and one aggregated over the plane, separate the same two failures along the same two axes. A member that takes one axis and not the other repairs one failure and keeps the other, which is why the family cannot be ordered on a single scale, and it is the reason the two moments are displayed separately here. Nothing in this statement is specific to ion channels: it holds for any state model whose observations are averages over a window, and it is the result of this paper that transfers furthest.

The two recursive members also differ in where on the plane their departures sit, and that matters as much as their size. Counting the fraction of cells that depart from one by at least 15%, parameter by parameter, `R` departs on 0.4375, 0.5625, 0.2232, 0.3125 and 0.0000 of the plane for k_{on}, k_{off}, i, N_{ch} and the noise level, against 0.0179, 0.0804, 0.0000, 0.0268 and 0.0000 for `IR`. The distortion of the instantaneous filter is spread across parameters and across the plane; the distortion of the boundary-conditioned filter is confined to the few-channel corner. Conditioning localises the error as well as shrinking it.

Instrumental noise moves the whole picture continuously, and it moves the window-ignoring members furthest. At $N_{\text{ch}} = 10$ the k_{off} distortion of least squares runs 11.79 at noise label 0.1, 1.14 at label 100 and 1.00 at label 1000; at $N_{\text{ch}} = 10^4$ the same sequence runs 12.97, then 2.54 at label 10^4, 1.15 at 10^5 and 1.00 at 10^6. The recursive instantaneous filter follows the same pattern about three decades lower, running 1.42, 1.03 at label 100 and 1.00 at label 1000 at $N_{\text{ch}} = 10^4$. The reason is that instrumental noise adds a variance that carries no gating memory, so as it grows it dilutes the temporal structure a window-ignoring likelihood fails to model, until there is nothing left to fail on. The channel count at which each member's error bar becomes honest therefore rises with the noise level, which is the crossover Figure 6 draws as a boundary.

Over the whole measured grid, counting every parameter at every cell, the reported uncertainty is within $\pm 15\%$ of the delivered one for 93% of the boundary-conditioned member's points (2116 of 2268, over 560 cells), 70% of the recursive instantaneous member's (684 of 980, over 238 cells), and 31% of the open-loop member's (277 of 896, over 224 cells); the one-endpoint member carried in the supplement reaches 20% (67 of 336). Least squares carries no entry in this comparison, because its distortion matrix has four dimensions against the likelihood members' six and the two counts are not commensurable (Methods). The coverage is honestly unequal and has to be quoted with the number: the boundary-conditioned member spans eleven channel counts from 5 to 10^4 across the twelve noise levels of the ladder (Methods), not on a full rectangle, and the others cover less.

The boundary-conditioned member is itself an approximation and fails in both directions. Over the whole measured grid its distortion runs from 0.645, at $N_{\text{ch}} = 10$, noise label 0.05, $\Delta \cdot k_{\text{off}} = 0.01$, on the channel-number direction, where it over-states its own uncertainty, to 1.701, at $N_{\text{ch}} = 5$, noise label 0.1, $\Delta \cdot k_{\text{off}} = 1$, on the closing rate, where it under-states it. Both excursions are in the few-channel corner, which is where the occupancy closure of the Theory section is expected to degrade.

The mechanism behind the flat floor is the split of the distortion into a per-sample part and a correlation part (Figure 4–figure supplement 1). On the closing rate at noise label 0.1, the per-sample part goes to one as channels are added for both recursive members, 1.09 to 0.998 for `R` and 1.10 to 1.007 for `IR` across the four decades, so the single-interval Gaussian approximation becomes exact for both at the same rate. What separates them is the correlation part, which rises for `R` from 1.35 to 1.47 and falls for `IR` from 1.24 to 1.00. The two members are equally faithful to a single interval, and they part company on how much temporal dependence they leave in the score, which is what conditioning on both interval endpoints removes.

The same memory is visible in the data, without any parameter. The integrated autocorrelation of the standardized residual, which counts how many intervals a likelihood effectively has for every interval it believes it has, reads 1.008 for `IR`, 1.30 for `R`, 3.93 for `NR` and 9.26 for `LSE` at noise label 0.1. The classical fit therefore behaves as though it had about nine times more independent observations than the recording supplies, which is the degrees-of-freedom cost of fitting correlated data as though it were independent.

Two features of the plane are easy to misread and belong in the text. The first is the left edge. Where the acquisition interval is far shorter than the channel's correlation time, successive samples carry redundant information, and a faithful likelihood has to register that redundancy, so a large correlation distortion at the shortest intervals is the correct cost of oversampling rather than a defect. The second is a single cell carried openly. At $N_{\mathrm{ch}} = 10^4$ and $\Delta \cdot k_{\mathrm{off}} = 1$ the amplitude directions of the recursive instantaneous filter are not identified: the Gaussian Fisher information is ill-conditioned there, so a displacement that costs almost no likelihood is amplified into a large parameter shift, and the reported bias runs off the scale of the map. Those cells are drawn grey. Their sign and their existence are real; their magnitude is a property of the conditioning and is not a physical bias.

The information budget per parameter

A calibrated member's reported covariance can be spent, and Figure 5 spends it. Only the boundary-conditioned member is shown, because the quantity is a covariance the member reports and only a member whose report has been checked can guide a design. The covariance used is the distortion-corrected one, so what is plotted is the precision the experiment achieves.

The quantity is a trade-off at a fixed channel budget. Plotting N_{ch} times the corrected variance of a parameter gives, up to the fixed budget constant, the variance of the estimate pooled over as many recordings as the budget allows. A flat line means the information is linear in channels, so it makes no difference whether the budget is spent as a few large patches or many small ones. A rising line means concentrating the budget into few large recordings yields less information.

The rates and the amplitude directions give opposite advice. Over the four decades of channel count at noise label 0.1, the pooled variance grows by a factor of 1.24 for the closing rate and 3.11 for the opening rate, against 288.19 for the unitary current, 60.50 for the channel number and 3236.85 for the instrumental noise level. For the rates, a channel budget can be concentrated at almost no cost. For the amplitude, the count and the noise level, the same budget spent as many small recordings gives a substantially better estimate. A noisier rig penalises the few-channel end, where the instrumental noise is large next to the gating variance, and it does so first in the rates, which are otherwise insensitive to how the budget is split. The acquisition interval runs the same way in every panel, with the pooled variance smallest at short intervals and growing toward the coarse end, mildly for the rates and more for the amplitude directions, which agrees with the calibration trade-off of Figure 4.

The scope of this figure is narrower than the trade-off it states. Everything here is the non-stationary protocol, a single concentration jump with a rising and a relaxing phase. The rates are recoverable from the shape of that deterministic transient, which is why they cost so little. Under a stationary protocol the mean current is flat, the transient carries no rate information, and the balance between the rates and the amplitude directions inverts. The advice below should not be carried across that boundary.

Where the calibrated member departs, and which moment carries it

Figure 5 resolves the corner where the boundary-conditioned member departs, on a scale that Figure 4 cannot show: there a factor of 1.5 is a faint tint, and here it needs an axis. Two things are measured on it.

The departure is two-sided within a single cell. At ten channels and $\tilde{S} = 0.005$ the reported error bar of the closing rate is too small by a factor of 1.62 at the shortest interval, while that of the channel number is too large by a factor of 0.65, so one likelihood is over-confident about a rate and conservative about a count at the same time and in the same recording. No scalar summary of the matrix expresses that, which is why the magnitude and the anisotropy are reported as a pair (Diagnostics).

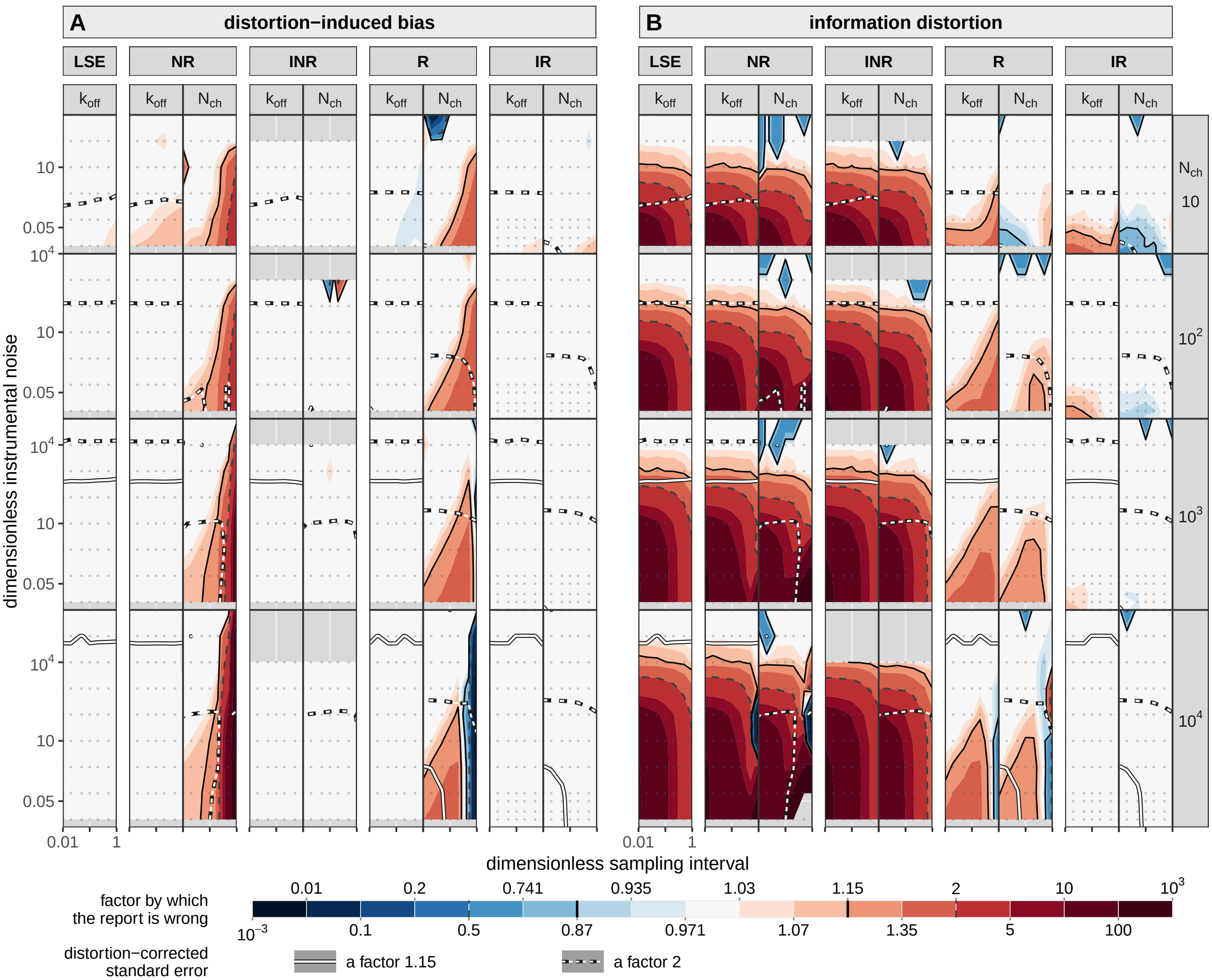

Figure 4. The design plane: how wrong each member's report is, in both moments, on one scale. Ten thousand recordings are simulated at every cell of the grid and scored by each member of the cost ladder. (A) The distortion-induced bias, evaluated at the simulation truth θ_{sim}: how far the estimate is displaced. (B) The information distortion, evaluated at the pooled optimum θ_{pool}, where the score vanishes by construction so a displaced gradient cannot contaminate it: how far the uncertainty a member reports is from the one it delivers. Columns nest twice, member and then parameter, k_{off} and N_{ch}. Least squares holds one column per half, because its four-parameter configuration fixes the unitary current and the noise level at the simulated values (Methods), so a channel-number comparison against it would not be like for like. Rows are the channel count, each given a height proportional to the range of instrumental noise the sweep reaches there, so a decade of noise is the same distance in every row. Inside a panel the axes are the sampling interval in units of τ and the dimensionless instrumental noise, and one decade is the same physical distance on both, so a field's shape can be read as a shape. Colour is shared by the two halves: the factor by which the report is wrong, centred on one, pale right and saturated wrong in either direction, blue conservative. What a factor means differs between them and the difference is not cosmetic: in (A) it is a factor on the parameter, so 2 means the estimate is out by a factor of two, while in (B) it is a ratio of variances, so 2 means the reported error bar is $\sqrt{2}$, about 40%, too narrow. The scale is clipped at $10^{\pm 3}$, which touches 22 of 3276 cells, twenty of them the open-loop member, where being wrong by 10^3 and by 10^{12} say the same thing. Lines carry two codes: colour says which quantity, dark hairlines cutting the colour field and white lines on a dark casing cutting the distortion-corrected standard error, which asks whether the parameter can be measured at all rather than whether its error bar is honest; dash says which criterion, solid a factor 1.15 and dashed a factor 2, the two thresholds used throughout. Both criteria are edges of the colour bands, which is why the dark pair needs no key of its own.

Figure 4—figure supplement 1. The distortion decomposed into a per-sample part and a correlation part, against channel count and resolved by instrumental noise. The per-sample parts of the two recursive members coincide and go to one as channels are added, while the correlation parts separate.

Figure 4—figure supplement 2. The sandwich prediction against the empirical multivariate distribution of

The two parts of the decomposition answer to the acquisition interval in opposite directions. Across that same cell the correlation part of the closing rate falls from 1.52 to 1.27 as the interval coarsens, while the per-sample part rises from 1.06 to 1.19: sampling more coarsely buys back the memory and pays for it in the per-sample non-Gaussianity. Both parts vanish by $\tilde{S} = 1$, where the instrumental noise has made the emission Gaussian enough that neither closure is strained.

Two limits travel with the figure. The family applies both of its Gaussian approximations at every step and the score sees only their product, so a departure measured here cannot be assigned to one of them; separating them needs the microscopic recursion, which is a companion paper's subject. And the grid is truncated where the effect grows: the departure of the closing rate is still accelerating at the lowest simulated noise, reading 1.22, 1.37 and 1.52 over the last three noise steps at ten channels, so the figure locates the onset and not the far side.

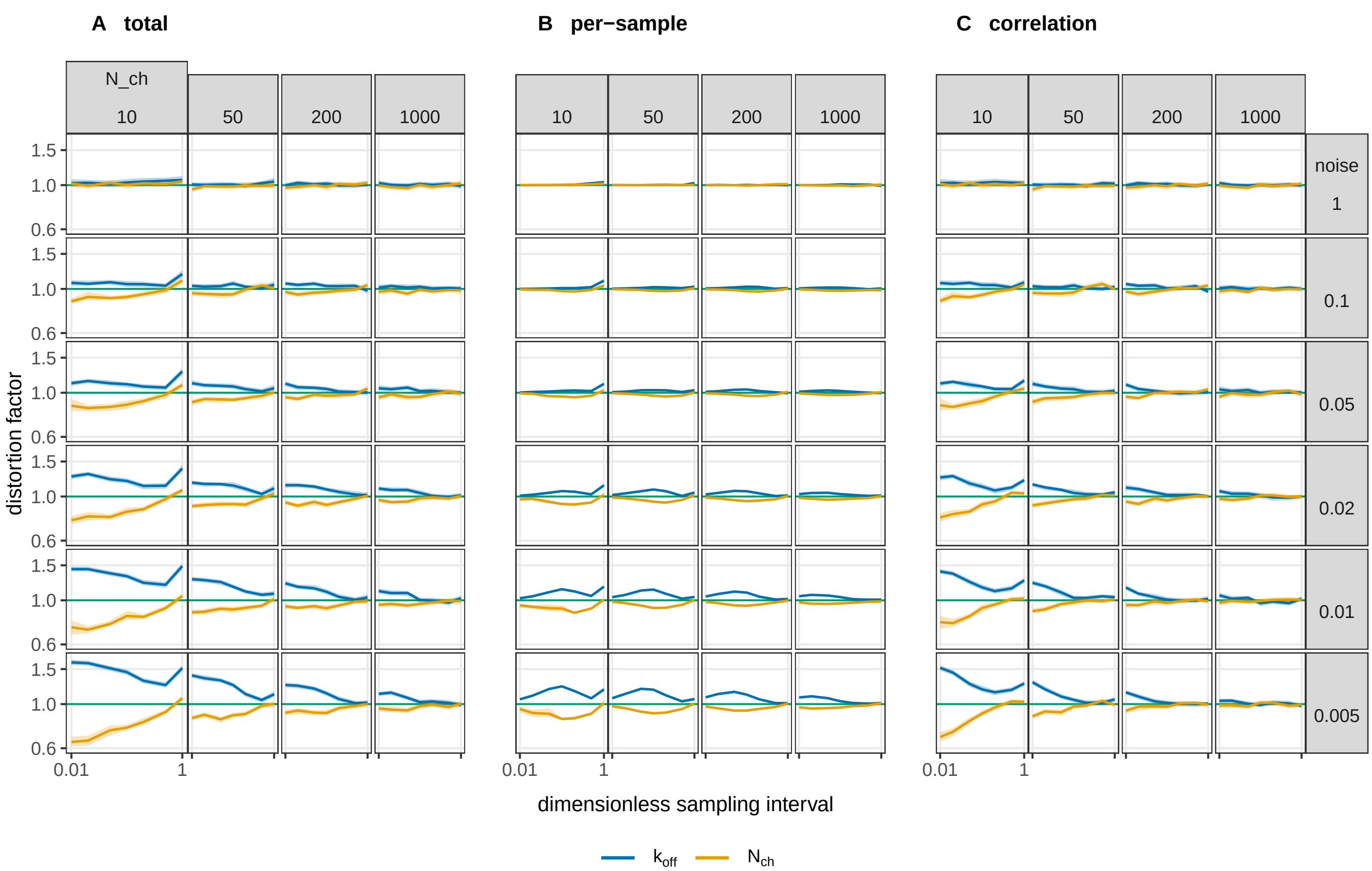

Figure 5. Where the boundary-conditioned filter stops reporting an honest error bar, and which of the two discarded moments is responsible. The same measurement as Figure 4, the Gaussian information distortion at the pooled optimum on the same runs, resolved on the corner where IR departs: on Figure 4's scale a factor of 1.5 is a faint tint and here it needs an axis. The three blocks are the distortion and the two parts it decomposes into: (A) the total, (B) the part carried by the per-sample non-Gaussianity of the score, and (C) the part carried by its correlation across intervals; their product reproduces the total to within 1% over the whole grid. Each block is a grid of the design plane in the orientation of Figures 4 and 6: the channel count increases to the right, the dimensionless noise $\tilde{S}$ increases upward, and inside every cell the horizontal axis is $\Delta \cdot k_{\mathrm{off}}$ from 0.01 to 1, shared by every panel and labelled once per block. The vertical axis is the distortion factor on a logarithmic scale, shared by all twelve columns so that they can be compared, with the green line at 1 marking a report that is neither over- nor under-confident. Colour is the parameter. Bands are 95% percentile bootstrap intervals over recordings, of median half-width 0.03. The two readings the figure supports, and the limits on them, are in the text.

A usage map for a macroscopic recording

Figure 6 collects the measurements into one plane, the channel number against the instrumental noise, at a single acquisition interval $\Delta \cdot k_{\mathrm{off}} = 0.1$. Unlike the earlier maps, its content is the boundaries: the measured cells are the input and the lines are the output. Four boundaries are drawn, and they are two questions asked of two methods. Two ask whether a method's reported error bar is honest, one for the boundary-conditioned likelihood and one for least squares, and are drawn where that method's information distortion crosses 1.15, a 7% error on the reported standard deviation. Two ask whether a parameter is reachable at all, one for the amplitude pair and one for the rates, and are drawn where the distortion-corrected standard error crosses a factor of two. The rate boundaries are taken from the least-squares arm and the amplitude boundaries from the likelihood, because the least-squares arm of this figure is the four-parameter configuration, which holds the unitary current fixed and so does not estimate it (Methods), while the two arms agree on the rates over the channel counts where both are measured and least squares is the arm with the high-noise coverage.

Read upward at a fixed channel count, the recording loses capabilities in a fixed order, which gives five regions. In region 0, below the lowest boundary, the Gaussian closure is itself misspecified, so the likelihood's own error bar is wrong and the recording is not a macroscopic problem. Region 1 delivers the unitary current, the channel number and an honest interval, while least squares reports an interval several times too narrow. In region 2 the amplitude pair is no longer separable and the rates alone survive, so the likelihood still buys an honest interval. In region 3 least squares is calibrated too and buys the same answer more cheaply. Above region 3 nothing useful is left. The boundary of region 0 is the one with a negative slope, measured at -0.56 in $\mathrm{d}\log_{10}\mathrm{noise}/\mathrm{d}\log_{10} N_{\mathrm{ch}}$, because instrumental noise makes the emission more Gaussian and so relaxes the occupancy closure while hurting everything else.

The boundary at which the classical error bar becomes honest is the one this paper predicts and then measures. In noise-label units it sits at 164 and 130 at $N_{\mathrm{ch}} = 10$, at 1.28×10^3 and 1.53×10^3 at 100, at 1.54×10^4 and 1.51×10^4 at 1000, and at 1.62×10^5 and 1.59×10^5 at 10^4, taking the channel-number and the closing-rate directions in that order; the dimensionless noise is one tenth of each. Fitted in logarithms, those crossings give slopes of 1.01 and 1.03, so the boundary runs as noise proportional to channel number. That is the crossover the mechanism predicts, since the gating variance grows with the channel count while the instrumental variance does not, so the ratio of the two is what decides whether a window-ignoring fit is honest. Here it is a measured slope rather than an assumed scaling.

The ordering of the boundaries is the result the map exists to state. There is no region of the measured plane in which the amplitude still carries information and the classical error bar is simultaneously trustworthy; the two boundaries are separated by one to three decades of noise everywhere measured. Whenever the fluctuations still say something about the unitary current or the channel number, a classical fit reports a confidence interval that is too narrow. The scope of that statement has to be stated with it: the classical arm here is least squares on the deterministic mean current with the unitary current held fixed, in its four-parameter configuration. It is not the variance-mean regression, this paper does not benchmark non-stationary fluctuation analysis, and nothing here should be read as a measured comparison against it.

The map also fixes where a parameter stops being reachable at any noise level. Minimising the corrected standard error over the whole noise sweep, the channel number is unreachable below about 17 channels and the opening rate below about 52, at this acquisition interval. The unitary current and the closing rate have no such floor within the simulated range.

The evidence is thinner than the picture, and the subsection says so rather than leaving it to a caption. The two boundaries at which least squares becomes honest, and the closure floor, are each fitted through four measured crossings. The channel count at which region 3 opens spans 120 to 1368 depending only on whether the strict or the loose criterion is used. The lower vertex, where the map closes from below, is extrapolated to N_{ch} between 3 and 9, below the simulated floor of ten channels, and the ten cells that would pin it have not been run. All the boundaries are drawn at one acquisition interval, with the ribbons showing where the same boundary moves as the interval is swept from 0.01 to 1 in units of τ. The map is a concept map, not a phase diagram. Its boundaries are level sets of continuous diagnostics, so a looser criterion moves each of them by up to a decade in noise without changing the layout.

Three recording configurations are placed on the same plane, as footprints rather than points: an excised outside-out patch, a whole cell, and a two-electrode voltage-clamped oocyte. Each footprint is the range of channel numbers the configuration expresses crossed with the range of instrumental noise it carries, converted into the sweep's own axis through the unitary current and the channel time constant, and clipped to the relaxation each configuration can resolve. In the sweep's label units the patch spans about 3×10^{-5} to 2.5, the whole cell about 10^{-4} to 20, and the oocyte about 8×10^{2} to 9×10^{6}. The placement is provisional and is offered as orientation. Typical instrumental noise per configuration is not published in a form that can be used directly, so the noise ranges are read off published current traces by a calibrated procedure, and the minimum resolvable current for two of the three configurations is a working value that has not been sourced.

Discussion

Over most of the design space the boundary-conditioned member reports an uncertainty close to the one it delivers, and where it does not, the departure is measured rather than assumed. Of the 2268 points at which the boundary-conditioned member was measured, one per parameter per design cell over 560 cells, spanning eleven channel counts from 5 to 10^4 and twelve instrumental noise levels with seven acquisition intervals resolved inside each cell, 2116 (93%) have a reported uncertainty within 15% of the one delivered. An experimenter who does not want to consult a map can use that member everywhere, and where its reported interval is the wrong width is the few-channel corner. Where it is wrong, the correction that repairs it is the one the diagnostic already supplies: at $N_{\mathrm{ch}} = 10$ its nominal 95% region covers 0.914 of the estimates under the covariance it reports and 0.947 to 0.949 under the sandwich, over a thousand fits in six parameters.

That member is an approximation and is not exempt from the failure this paper is about. Over the same grid its distortion runs from 0.645, at $N_{\mathrm{ch}} = 10$ with noise label 0.05 and $\Delta{\cdot}k_{\mathrm{off}} = 0.01$ on the channel-number direction, where it over-states its own uncertainty, to 1.701, at $N_{\mathrm{ch}} = 5$ with noise label 0.1 and $\Delta \cdot k_{\mathrm{off}} = 1$ on the closing rate, where it under-states it. The departure is two-sided, and that is how it should be read: the reported interval moves in both directions as the channel count falls, and the approximation does not become safe by erring conservatively on average. Both excursions sit where the open population is a small integer count whose distribution is visibly discrete, which is where the occupancy Gaussian, the first of the family's two closures, is expected to degrade. No channel count is named as the point at which that closure becomes misspecified, and the reason is a limit of this design rather than a gap in the measurement: with the open probability held at one half throughout, a channel count and an open-channel count differ by a fixed factor of two that nothing here can separate, so a threshold stated in channels would be stated in the wrong unit. What is measured instead is where each parameter stops being reachable at all, below 17 channels for the channel count and below 52 for the opening rate. The boundary also moves with the instrumental noise, downward as the noise rises, because instrumental variance makes the emission more Gaussian and so relaxes the closure while hurting everything else.

A second boundary of the family's validity is argued here from its premise and has not been measured. Every member above least squares assumes that the instrumental contribution to a recorded interval is Gaussian and white, entering the predictive variance as one additive constant and leaving no memory between intervals. Recordings depart from that description, through the two-level switching of a trapped charge in the headstage and through the flicker component of the amplifier, and the discrepancy literature for models of this class already argues that unmodelled structure of exactly this kind, rather than independent Gaussian error, is what the residuals of a

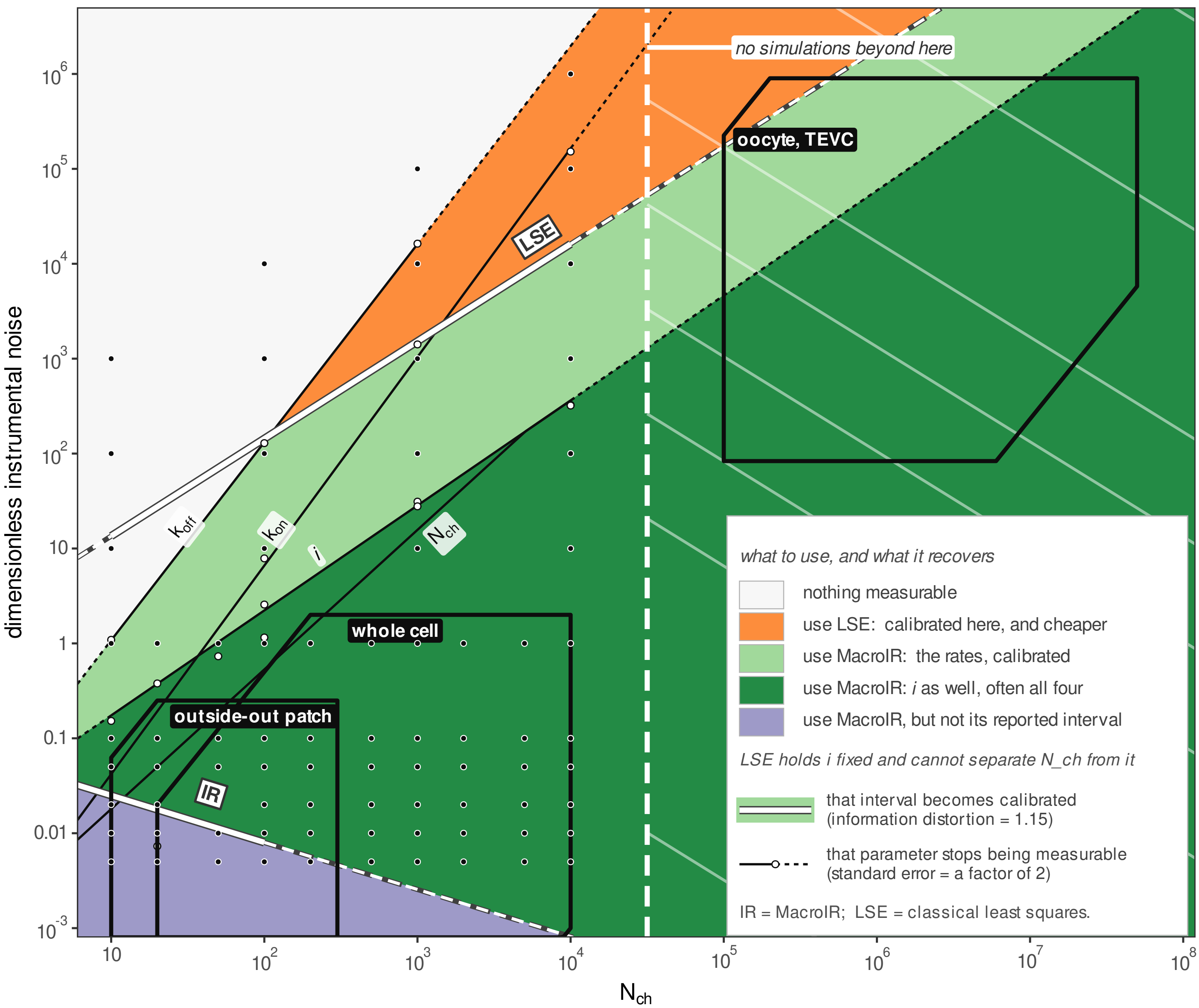

Figure 6. What a macroscopic recording can be asked for: a usage map over channel number and instrumental noise. The plane is the design space, the number of channels in the patch against the instrumental noise label, both on logarithmic scales, at one acquisition interval $\Delta \cdot k_{\text{off}} = 0.1$. Four boundaries divide it, and they are two questions asked of two methods. Colour is the parameter the boundary is about. Line type is the question: solid asks whether the parameter can be measured at all, drawn where the distortion-corrected standard error crosses a factor of two; dotted asks whether the reported error bar of least squares is honest, drawn where its information distortion crosses 1.15; dashed asks the same of the likelihood's own error bar, and is the boundary below which the Gaussian closure is misspecified. Ribbons are the same boundary swept over acquisition intervals from 0.01 to 1 in units of τ, so a ribbon says where a reader's own sampling rate puts the boundary and is not an error bar. Dots are the measured crossings and the line through them is their fitted power law, drawn faint where it leaves the simulated range, which is marked. Vertical strips mark the two vertices, each a range set by the strict and the loose criterion. The boundaries partition the plane into five regions, filled and keyed by what a recording in each can be asked for, and the three labelled boxes mark where common preparations sit; the regions are named and read in order in the text. The map is a concept map, not a phase diagram. Its boundaries are level sets of continuous diagnostics, so a looser criterion moves each of them by up to a decade in noise without changing the layout.

fitted electrophysiology model carry ***Lei et al.*** (***2020***). Where such a departure is large, the diagnostic would report a distortion that belongs to the noise model rather than to the gating closure, and the score alone does not separate the two. We have not run it. The simulator emits Gaussian white instrumental noise only, no cell with any other noise process exists in the sweep, and nothing in this paper measures the size of the effect. This is an expectation from the closure's premise rather than a result, and the way to settle it is to simulate the alternative noise process and repeat the same measurement.

The same amplifier stage carries a second departure of the same shape, and this one we adopt knowingly. A recording is low-pass filtered before it is digitised, normally by a Bessel filter, so the measurement kernel of a real experiment is that filter's impulse response, while the observable used here, in the simulator and in every likelihood alike, is the uniform average over the acquisition window (Theory). The uniform window is the first approximation to that kernel and it is the one that makes the family well posed, since it keeps all of the weight of a recorded value inside a single interval, which is the interval whose two ends the calibrated member conditions on. A Bessel kernel moves both premises of this paragraph at once: it spreads weight across interval boundaries, so successive recorded values are correlated through the signal itself, and it colours η_t. Neither effect is measured here. Extending the recursion to a general kernel with its coloured noise, and then re-running this same diagnostic against it, is the natural next step, and until it is taken the numbers above describe uniformly averaged recordings.

What buys the calibration is conditioning at both ends of the acquisition interval. Conditioning on the start alone does not buy it, and at the reference cell it costs. The one-endpoint member and the boundary-conditioned member assign the same total variance to the observed interval average, from the same state, and they book it differently: the spread of the interval mean across end states enters the one-endpoint member as observation noise, while the boundary-conditioned member carries it as a covariance between the interval current and the occupancy at the end of the window, which is the numerator of the Kalman gain (Eq. 12). The whole distance between the two therefore lies in the update. On the ellipse-area measure at $N_{\mathrm{ch}} = 100$, noise label 0.1 and $\Delta \cdot k_{\mathrm{off}} = 0.1$, the one-endpoint member reads 1.63 against 1.01 for the boundary-conditioned one, and it is also worse than the instantaneous recursive filter at 1.17, so the intermediate rung does not interpolate between the two ends of the step it sits in.

A supplement takes that step in its two algebraic halves and confirms a prediction that was registered before the run. The variance-recursive member removes the end-state spread from the observation variance without restoring it in the gain, and the prediction was that its reported uncertainty would come out worse than the one-endpoint member's, over-confident and more so. It does. The reading carries one qualification the algebra makes obligatory: the predictive variance divides the gain, so altering the variance form also moves the update, and the two halves are not a clean separation into a variance step and a gain step. This is a supplement result and the body's claim does not rest on it.

The diagnosis is Milescu's, now measured. ***Milescu et al.*** (***2005***) named the local time correlation of the current as the dominant error of estimates that do not filter, and that correlation is what the distortion decomposition isolates. Interval by interval every rung reports its information correctly; the miscalibration appears in how the intervals accumulate. It is visible directly in the autocorrelation of the score, which at lag one on the closing rate is -0.0043 ± 0.0043 for `IR`, indistinguishable from zero, against 0.191 ± 0.004 for `R`, 0.864 ± 0.0019 for `NR` and 0.856 ± 0.0022 for `LSE` (six-parameter configuration, Methods). Conditioning on both interval ends leaves a white score. A likelihood member that skips an endpoint reads sequential structure as independent evidence, which is how a per-interval identity that holds everywhere can accumulate into an information ratio off by more than an order of magnitude. The accumulated ratio of least squares is not read this way: its information presumes one constant residual variance, which the simulator violates, so its factor of fifteen measures the size of that break rather than a failure of accumulation, and it sits above the calibrated value at the per-step level too (Diagnostics).

Del Core and Mirams (***2025***) argue against filtering on cost, since the moment integration and the correction step run between every pair of samples. The cost is real, and what it buys is now quantified in units of misreported uncertainty. Where the instrumental noise is large next to the gating variance the cheaper members recover: at $N_{\mathrm{ch}} = 10^4$ the closing-rate distortion of the instantaneous recursive filter falls from 1.42 at noise label 0.1 to 1.03 at label 100 and 1.00 at label 1000, and least squares follows the same path about three decades higher in noise. In that part of the plane the filter's expense is hard to justify on calibration alone. Toward few channels and short intervals the correlation part of the distortion grows, and the degrees-of-freedom count a non-conditioned fit uses is wrong by a factor that no compute saving offsets: at noise label 0.1 the integrated autocorrelation of the standardized residual is 9.26 for least squares and 3.93 for the open-loop member, against 1.008 for the boundary-conditioned filter, so the classical fit behaves as though it had about nine times more independent observations than the recording supplies. The first moment is a separate charge on the same trade. The instantaneous filter carries a bias in the channel number of 35 to 40% that does not shrink over four decades of channel count, and no error-bar correction touches it. The map states both halves, so a reader can locate a recording on it and see which side of the trade it sits on.

One result speaks to how a recording should be spent. Once the agonist is removed and the open population relaxes, a member that has conditioned on the past gains nothing further about the channel number or the opening rate, whose information reaches the numerical floor within six intervals of the washout, while the closing rate and the unitary current stay informative through the decay by different routes, the first through its shape and the second through the fluctuation of the channels still open (Figure 3–figure supplement 1). The shape does not hold for every rung, and that is itself the finding: the non-recursive members keep about a sixth of their pulse information about the channel number through the washout against under one per cent for the recursive ones (Results), so this is a statement about conditioning and not about the observable. It answers a question ***Del Core and Mirams*** (***2025***) leave open. For the bench it says that the transient carries what the decay cannot: extending a record after the agonist is removed adds data without adding information about N_{ch} or k_{on}, so an experiment that wants those two should spend its samples on the jump and its rise, while an experiment that wants the closing rate or the unitary current is served by the decay as well.

The ordering of the boundaries on the map is what an experimenter can act on. There is no region of the measured plane in which the fluctuations still carry information about the unitary current or the channel number and the classical error bar is simultaneously trustworthy; the two boundaries are separated by one to three decades of noise everywhere measured, and the boundary at which the classical interval becomes honest runs with fitted log-log slopes of 1.01 and 1.03, so it scales as instrumental noise proportional to channel number. Each of those boundaries is fitted through four measured crossings at one acquisition interval. The map is a concept map, not a phase diagram. Its boundaries are level sets of continuous diagnostics, so a looser criterion moves each of them by up to a decade in noise without changing the layout. That scaling was predicted from the mechanism, since the gating variance grows with the channel count while the instrumental variance does not, and it is then measured as a slope. The concession that travels with it is that least squares recovers the rates, with a rate bias at or near zero over the same plane; what it gets wrong is how well it says it recovered them. The scope of the comparison travels with it too. The classical arm of the design-plane and map results is the four-parameter configuration of Methods, least squares on the deterministic mean current with the unitary current held fixed; the time-resolved comparison uses the six-parameter configuration, and no least-squares quantity is carried across the two. This paper does not benchmark non-stationary fluctuation analysis, no variance-mean regression is run anywhere in it, and nothing above is a measured comparison against that method.

As an algorithm, MacroIR is an integrated-measurement augmented Kalman filter, a device long known in control and target tracking ***Zadrozny*** (***1988***), and its predictions coincide with that construction to about 10^{-8}. What is specific here is the realization for an exact continuous-time Markov chain of channel occupancies, its use at the populations simulated in this paper, from 5 to 10^4 channels, where the single-molecule integrated filters do not reach, and the measurement of what the cheaper members cost. The device is conceded; the continuous-time realization ***Moffatt and Pierdominici-Sottile*** (***2025***) and the validity map are the contribution, and the recursive lineage they extend is ***Moffatt*** (***2007***) and ***Münch et al.*** (***2022***).

The full distortion matrix is reported, rather than a scalar summary, because it is the input to the next question. Near the maximum the same matrix propagates into the Bayesian evidence through a volume term $\frac{1}{2}\log\det\mathbf{C}$ and an effective-sample rescaling $\alpha^\star = p/\operatorname{tr}\mathbf{C}$, so whether model comparison under an approximate likelihood ranks mechanisms correctly is decided by the object measured here. That correction is derived elsewhere and is stated only as motivation. It is one bridge of a programme that assembles exact simulation, an approximate likelihood and its correction into inference on real recordings ***Moffatt and Pierdominici-Sottile*** (***2025***).

A reader who accepts the miscalibration can still ask what it would cost to repair it without changing likelihood, and the alternatives answer different questions. The standard error across independent fits to separate patches is free, since the fits already exist, but it describes the uncertainty of a population mean across patches, mixes statistical with biological variability, and leaves any bias in place while shrinking as the square root of the number of patches. Resampling whole sweeps and refitting is valid and correctly preserves the within-sweep correlation, and it gives an honest interval around whatever the estimator converges to while saying nothing about whether that estimator is the right one. Resampling residuals is not valid here, because independent resampling destroys the correlation that is the problem. A parametric bootstrap assumes the model one was trying to check. A block bootstrap needs a block length well above the correlation time, so it needs that correlation time measured before it can be configured. Simulating from the fitted model and forming the sandwich answers the same question the diagnostic does, and its cost is set by the precision needed on a covariance rather than by the model, which puts it orders of magnitude above a single evaluation and still leaves it simulating from the model one wanted to verify.

Where exchangeable sweeps are plentiful, the classical route is strong and should be said to be. Non-stationary fluctuation analysis returns the channel number and the unitary current with no likelihood at all, and a sweep-level bootstrap returns honest intervals with no model, and between them they cover much of routine electrophysiology. The case for a filter is strongest where that route is weakest: few sweeps, rundown, conditions that are not exchangeable, a single precious recording, and an experiment that wants kinetics and amplitudes jointly from the same record.

What this paper does not settle

The broad question, when to prefer one of these methods over another, cannot be closed with the elements assembled here, and the honest move is to say which parts of it are missing rather than to answer it badly. Four of them depend on facts about the reader's experiment that no simulation study supplies. The cost comparison in general depends on the number of states, the implementation and the optimiser, and this paper has one model, one implementation and one machine. Whether a sweep-level bootstrap suffices depends on how many sweeps a preparation tolerates and whether they are exchangeable, which is a rundown question. How much precision is lost by fitting with least squares is measured here for this model, and whether that loss matters is the reader's judgement about their own question. How any of this composes with non-stationary fluctuation analysis as a complete workflow is not addressed.

The fifth is about the diagnostic itself and bounds what it verifies. The comparison here is between two independent implementations, the likelihood and the exact simulator, and everything downstream of the point at which they separate is tested by it: the propagation, the update, the emission arithmetic, the accumulation. Upstream of that point they share one specification, the rate matrix built from the parameters, the conductance vector, the protocol and the sampling times, and a diagnostic built on their agreement cannot see an error there. For a two-state scheme that specification is small enough to be read and checked by eye, and it is written out in Methods for exactly that reason. The general remedy is different in kind: verify each object against the theorems it must satisfy, the semigroup and Chapman-Kolmogorov identities for the propagator, the stationarity condition, the normalisation of the occupancy, and the moment identities of the emission. That is a piece of work with its own design and it is not undertaken here. The sixth is the scope declared in the paragraph below.

What this changes is what the paper hands over. The reader supplies their sweep count, their rundown, their model size and their tolerance, and decides; what they cannot measure for themselves, and what is supplied here, is whether a reported interval is the interval that is delivered.

What produces the boundary-conditioned member's own departure is not known, and the shape of the ignorance is worth stating because it is sharper than a general disclaimer. The decomposition puts the departure in the term that accumulates across intervals rather than in the per-sample one: the member's per-sample non-Gaussianity stays close to one where its total does not. Five candidate criteria for where the closure should fail were derived and measured against the observed boundary, and every one of them was a criterion on the per-sample cumulants of the occupancy, so every one of them failed to locate it. What survives from that exercise is a bound and not a position: occupancy non-normality alone cannot produce a factor-two distortion anywhere on the measured grid. The measured channel-count exponent of the boundary is -0.25, which matches neither the -0.5 a skewness mechanism would give nor the -1.0 of a kurtosis one, and it is not a clean power law. Stated as a conjecture and in those words: the limits behave like a balance that has to come out just right, several quantities moving together in a way that leaves the departure small over four decades, which is the signature of a conserved quantity or a comparable invariant that has not been identified here.

The perimeter of these claims is a model, and it is declared as one. The scheme is two states with the open probability held at 0.5; the protocol is a single concentration jump; the data are simulated, because a diagnostic that compares what a likelihood delivers against what it reports needs a known truth to mean anything; and the inference is likelihood only, with no prior and no evidence computation. The two-state scheme isolates the closure's own error with no confounding from scheme misspecification, and the fixed open probability means that a channel count and an opening count differ throughout by a factor of two that no measurement in this paper can settle. Richer schemes, the stationary regime, the few-channel boundary, the filtered measurement kernel and experimental data are named companions rather than omissions. The machinery itself is not specific to ion channels: simulate a process exactly, test the score against the simulation, sandwich the Fisher information, read off the distortion. It applies to any approximate likelihood for a process that can be simulated exactly but whose likelihood cannot be written down, which is most of stochastic biophysics.

Materials and methods

Minimal model and simulation

We work with a two-state channel scheme, closed (C) and open (O), the smallest scheme in which the acquisition-averaging effect we study is present. The closed-to-open transition is agonist dependent with association constant k_{on} multiplying the agonist concentration; the open-to-closed transition is agonist independent with dissociation rate k_{off}. Only the open state conducts, and the code applies a sign flip so the single-channel current is inward. At the start of every recording all channels are closed, so the occupancy is $(1, 0)$.

The scheme carries six parameters, all estimated in base-ten logarithmic coordinates. The values used to generate the data reported here, read from the production run scripts rather than from the parameter tables in the repository (the tables hold stale values the figures never used), are $k_{\mathrm{on}} = 10\ \mu\mathrm{M}^{-1}\,\mathrm{s}^{-1}$, $k_{\mathrm{off}} = 100\ \mathrm{s}^{-1}$, a unitary conductance g set to 1 (the engine parameter `unitary_current`, written i where the noise axis is defined below), and a current baseline b set to 1 (both in units of g), while the white-noise level and the mean channel number were swept (below). Current is expressed throughout in units of the unitary conductance g, which corresponds physically to a single-channel current of roughly $1\ \mathrm{pA}$. The dissociation time $\tau = 1/k_{\mathrm{off}} = 10\ \mathrm{ms}$ sets the natural time scale that normalises the acquisition-interval axis. The relaxation at the agonist condition is faster, with rate $k_{\mathrm{on}}[A] + k_{\mathrm{off}} = 200\ \mathrm{s}^{-1}$; the axis uses $\tau = 1/k_{\mathrm{off}}$ by convention.

The emission model is white instrumental noise only. Both the pink (one-over-f) and the proportional noise channels are set to zero, so the term the reader may expect from flicker noise is genuinely absent. For a measurement interval t of duration Δ_t, the predicted current has mean $\bar{y}_t = N_{\mathrm{ch}}\, \mu_t \cdot \gamma_t + b$ and variance

$$\sigma_t^2 \;=\; e_t \;+\; N_{\mathrm{ch}}\, v_t, \qquad e_t \;=\; \mathrm{Current\_Noise}\, \frac{f_s}{n_{\mathrm{samp},t}} \;=\; \frac{\mathrm{Current\_Noise}}{\Delta_t},$$

where N_{ch} is the channel number, μ_t the occupancy mean, γ_t the interval-averaged conductance, f_s the raw sampling rate, $n_{\mathrm{samp},t}$ the number of raw samples averaged into the interval, e_t the white instrumental variance over the interval, and $N_{\mathrm{ch}}\, v_t$ the channel-gating variance. The white variance e_t falls as the interval lengthens, because more raw samples are averaged into each recorded value. The gating variance v_t is the single-interval conductance variance, whose form differs across the algorithm family and is derived in the Theory section.

The stimulus is a single concentration jump in three segments, a pre-pulse at $0\ \mu\mathrm{M}$, an agonist step at $10\ \mu\mathrm{M}$, and a post-pulse at $0\ \mu\mathrm{M}$, with raw sampling at $f_s = 50\ \mathrm{kHz}$ throughout. Each recorded measurement is the mean of $n_{\mathrm{samp},t}$ consecutive raw samples, so the averaging window is fixed at acquisition. That window is the acquisition interval the whole study is about, and its length is set by the number of raw samples entering each recorded value. The averaging is uniform over the window and no analogue filter stage is simulated, so a recorded value is a boxcar average of the trajectory, which is the observable every likelihood in the family is written for (Theory). Recordings from a real amplifier are Bessel filtered instead, and that kernel is outside the scope of this paper (Discussion).

Ground truth is generated by exact simulation of the continuous-time Markov chain (CTMC) by uniformization, so the interval average of the conductance is realised from an actual sampled trajectory rather than from any moment-closure approximation. Uniformization draws the transition times themselves, so there is no within-interval discretisation to choose and none to justify. The production scripts do pass a `number_of_substeps` argument, and on this branch it has no effect: the simulation parameters are built from the algorithm name alone, and the substep count they carry is zero. This is what licenses treating the simulation as the reference against which the Gaussian likelihood approximations are judged: the approximations are the object under test, and the simulator supplies the exact answer they are compared to. Every maximum-likelihood grid cell reported here, the least-squares arm included, used 10^4 recordings, with one exception: the cell that carries the coverage result ($N_{\mathrm{ch}} = 10$, noise label 0.05, boundary-conditioned member) was run at 10^5, because coverage of a nominal 95% region in six parameters needs a thousand independent fits and those fits are taken at group size 100, so 10^5 recordings are required to produce them. The time-resolved run used $1{,}000$ recordings; the single-recording illustration (Figure 1) was drawn by the substep sampler instead, one 20-channel trace at 250 substeps per interval, and does not enter the quantitative comparison.

The simulation seed was set to 0, which in this engine is the sentinel for a random seed drawn from `std::random_device`, and the resolved seed was never logged. The deposited ensembles are therefore statistically equivalent to, but not bit-for-bit reproducible from, a rerun of the same script. Because each ensemble is large ($1{,}000$ to $10{,}000$ recordings), the reported statistics are stable; the Monte-Carlo standard here is statistical equivalence, not bit identity.

The likelihood family and its members

The members compared here answer one question first, then a second. The first question is whether the gating fluctuations are modelled at all. Classical nonlinear least squares answers no: it fits the deterministic mean current and treats every departure from it as independent measurement error. The macroscopic likelihoods answer yes, and among them the second question is how much of the acquisition interval the interval-averaged conductance is conditioned on. Least squares is the bottom rung of that cost ladder and the comparison anchor of this paper.

Least squares is selected by the family flag *family approximation* $= 2$, its data key is `nonlinearsqr`, and it runs non-recursive with the averaging flag at 1; the variance flag is accepted and ignored on this branch. It sits off the lattice of endpoint and recursion flags that organises the macroscopic members, so it carries no compositional name. The macroscopic members all take *family approximation* $= 0$ and are one filter run under flags that set how the interval-averaged conductance is conditioned. The *recursive* flag sets whether the occupancy covariance is propagated between intervals. The *averaging* flag $\mathrm{av} \in \{0, 1, 2\}$ counts the number of interval endpoints the hidden state is conditioned on. A third flag, the *variance form*, distinguishes the two members that condition on the interval start: MR predicts the interval variance with the total per-start-state conductance variance, while VR removes the between-endpoint part, keeping MR's mean and gain.

Member	Data key	Recursive	av	Variance	Conditioning of the conductance
LSE	`nonlinearsqr`	no	1	n/a	none; the mean current only
NR	`macro_NR`	no	0	n/a	instantaneous; the acquisition averaging is ignored
INR	`macro_INR`	no	1	total	interval mean, without recursion; the endpoints are not separable
R	`macro_R`	yes	0	n/a	instantaneous; the acquisition averaging is ignored
MR	`macro_MR`	yes	1	total	interval mean given the start state
VR	`macro_VR`	yes	1	residual	MR's mean and gain, with the between-endpoint variance removed
IR	`macro_IR`	yes	2	residual	interval mean given both boundary states; the endpoints enter the gain

Table 2. The members as flags. LSE takes *family approximation* $= 2$ and every macroscopic member takes *family approximation* $= 0$. Reading the endpoint flag as a ladder gives no endpoints (NR, R), one endpoint (MR, VR), then both boundary states (IR); INR sits outside that ordering, because without a recursive update the boundary state is unobservable and $\mathrm{av} = 1$ and $\mathrm{av} = 2$ describe the same method. The variance column reports the form in force rather than the dispatched flag: IR is dispatched with the total flag, but at $\mathrm{av} = 2$ the residual form is used regardless of it, because the between-endpoint spread is already carried by the boundary term. Even so, MR and IR predict the same total interval variance and differ only in the gain: the two decompositions rearrange into each other (Appendix 1). VR shares MR's gain but removes the between-endpoint part of the variance, which isolates the variance change from the gain change. LSE and the macroscopic members are written by different producers, so their output files carry different prefixes, `figure_3_LSE_` and `figure_3_G_`. INR is the member published as MacroINR in ***Moffatt and Pierdominici-Sottile*** (***2025***), where it served as the control against which the boundary-conditioned member was compared; it is reported here only in the supplement.

Table 2 lists the members. Reading the endpoint flag as a ladder, the macroscopic members condition the conductance on progressively more of the interval: no endpoints, one endpoint, then both boundary states, with the exact trajectory (what the simulator supplies) as the unreachable top rung. `IR` is the top implemented rung: it conditions on the pair of channel states at the two ends of the interval, which we call the boundary state, and marginalises the interior analytically. It is the likelihood introduced for the P2X2 receptor (***Moffatt and Pierdominici-Sottile, 2025***), here placed as one member of the completed grid, and it is verified in the Theory section to coincide with a time-integrated Kalman filter. The construction of the interval-averaged mean and variance, the boundary-state covariance, and the Kalman-like update are derived in the Theory section and its supplement; we do not restate them here.

Two reductions are worth stating because they anchor the family to prior work. Setting the boundary-conditioned conductance variance to zero in the recursive, no-endpoint member (R) recovers the macroscopic-fluctuation update of ***Moffatt*** (***2007***), so R is the recursive ancestor already in print. The single structural difference between the M and I families is that the one-endpoint members drop the boundary cross-covariance term $N_{\mathrm{ch}}\,\widetilde{\boldsymbol{\gamma}^{\top}\boldsymbol{\Sigma}\boldsymbol{\gamma}}$ that IR retains, the tilde being the boundary-pair contraction defined at Eq. 11. Every member is written out on one template in Appendix 1.

Throughout the reported runs the remaining build flags were held fixed: *taylor variance correction* off, *micro approximation* off, and the *variance approximation* on. The V in VR names the *variance form* flag, set to 1, and has no connection to the *taylor variance correction* flag, which is off in every run reported here; the Taylor-corrected variants MRT and IRT are not part of this paper. Because the Taylor variance correction is off, the recursive one-endpoint member that runs is MacroMR and not MacroMRT.

The filter carries two numerical safeguards, present because the code runs them and because anyone reimplementing the filter meets the same constraint on day one. Without them the recursive members return no value at all at the smaller channel counts, so they are part of the method rather than housekeeping.

The first holds the occupancy mean on the probability simplex. An undamped Kalman-like update can drive entries of $\boldsymbol{\mu}$ negative or above one, so the mean step is scaled by a trust coefficient $\alpha_{\mu} \in (0, 1]$, the largest scaling that keeps the posterior on the simplex. The covariance is treated separately and takes its full step, since the rank-one down-date of Eq. 13 is positive semidefinite by construction; the two trusts are not tied to a common factor. The form of the damping matters here in a way it would not in an ordinary filter. This paper differentiates the log-likelihood with respect to $\boldsymbol{\theta}$, so a coefficient that is merely continuous is not enough: a hard $\alpha = \min(1,\ c\,\alpha_p)$ leaves a step in $\partial\alpha/\partial\boldsymbol{\theta}$ at the boundary and pumps variance into the score whenever the system sits near it, and a discontinuous form leaves a delta. The coefficient is therefore taken in the smooth form $\alpha = \frac{1}{2}\big(1 + c\,\alpha_p - \sqrt{(1 - c\,\alpha_p)^2 + \epsilon^2}\big)$, a softened minimum of 1 and $c\,\alpha_p$ that is infinitely differentiable in $\boldsymbol{\theta}$ everywhere. Here α_p is the largest scaling that keeps the posterior mean on the simplex, $c = 0.9$ is a safety factor applied when the constraint binds, leaving a ten per cent margin against the boundary, and $\epsilon = 10^{-4}$ sets the width of the smoothing band, so the residual bias of α below the hard minimum is at most $\epsilon/2$, which is three orders of magnitude smaller than that margin. Reimplementing this with a hard minimum reproduces the mean current and corrupts every quantity this paper measures.

The second is a pair of constants: a binomial-count floor of 5, below which the expected count in a state is too small for the Gaussian treatment of that state and the member falls back to its non-recursive form, and a minimum occupancy probability of 10^{-12}.

The two least-squares configurations

The least-squares arm runs in two configurations, and they are different statistical models. Both select *family approximation* $= 2$, both fit data from the same simulator, and both minimise the same sum of squares. They differ in which parameters are free, so no quantity computed under one may be compared against the same quantity computed under the other.

The residual scale is not a free parameter of either and is not given a value. It is integrated out under a scale-invariant prior, so with $\mathrm{SSE} = \sum_i (y_i - \mu_i(\boldsymbol{\theta}))^2$ over the n finite recorded values the objective is the marginal

$$\log L(\boldsymbol{\theta}) \;=\; \log\Gamma\!\left(\tfrac{n}{2}\right) \;-\; \tfrac{n}{2}\log\pi \;-\; \tfrac{n}{2}\log\mathrm{SSE}(\boldsymbol{\theta}), \tag{18}$$

whose maximiser is the ordinary least-squares one, since only the last term depends on $\boldsymbol{\theta}$. What the marginalisation changes is everything downstream of the point estimate. The score is the derivative of Eq. 18 and carries the factor n/SSE rather than a $1/\sigma^2$ that would have to be supplied, and the information the fit reports is

$$\mathbf{H}_{\mathrm{LSE}} \;=\; \frac{n}{\mathrm{SSE}} \sum_i \frac{\partial\mu_i}{\partial\boldsymbol{\theta}}\,\frac{\partial\mu_i}{\partial\boldsymbol{\theta}}^{\top}, \tag{19}$$

the Gauss-Newton curvature with $\hat{\sigma}^2 = \mathrm{SSE}/n$ entering as a plug-in prefactor and not as a coordinate. There is therefore no $\partial\sigma^2/\partial\boldsymbol{\theta}$ term to keep or drop: the scale direction is not in the parameter vector at all, which is also why the classical arm carries no score in the noise direction and why its mean squared standardized residual is pinned at one by construction.

The six-parameter configuration is `ops/local/figure_3_time_LSE.macroir`, and it feeds the time-resolved figure (Figure 3). All six parameters are free in base-ten logarithmic coordinates: k_{on}, k_{off}, the unitary current, `Current_Noise`, the current baseline and the mean channel number. The script's own header records why. The time-resolved figures accumulate the per-interval Fisher information and never invert it, so the flat directions of the least-squares fit are harmless there; and keeping six free preserves the parameter-index alignment with the macroscopic dumps that the figure code joins against, whereas fixing two would emit four indices against the macroscopic files' six and misalign them without any visible error. Two flat directions are then visible rather than hidden. `Current_Noise` has an identically zero score row and an identically zero Fisher row, because the least-squares mean carries no noise term; and the unitary current and the mean channel number form a rank-one degenerate block along the amplitude ridge $N_{\mathrm{ch}}\, i$, so the unitary current is free in this configuration without being separately identified.

The four-parameter configuration is `ops/local/figure_4_LSE.macroir`, which produces the grid cells, together with `ops/local/figure_3_mle_LSE.macroir`, which is the same model run with the per-group estimate cloud retained for the few cells that need one. Both fix the unitary current at 1 and `Current_Noise` at the cell value, leaving four free parameters: k_{on}, k_{off}, the current baseline and the mean channel number. Fixing those two is forced rather than chosen: they are the flat directions of the previous paragraph, and with them free the parameter Fisher information is singular, so no maximum-likelihood covariance exists. Both fixed values are the values the recordings were simulated from. The script builds one parameter vector and passes it both to the simulator and to the fit as its reference point, so the fixed value and the truth are the same number by construction. Least squares is therefore not handicapped in this configuration but helped: two of the six parameters are supplied to it exactly right, while the likelihood members estimate all six. Where it is nonetheless the arm whose reported uncertainty is furthest from the delivered one, that failure is not a lack of information about those two. This is the configuration behind every least-squares cell of Figures 4 and 6. Its parameter indices are $0 = k_{\mathrm{on}}$, $1 = k_{\mathrm{off}}$, $2 =$ current baseline, $3 =$ mean channel number. There is no unitary-current index in these files, and index 2 is the baseline.

The consequence is carried into the Results. A score, a Fisher information, a distortion matrix or a log-likelihood computed under six free parameters is a different object from the one computed under four, and the two distortion matrices differ in dimension, 6×6 against 4×4, so any scalar that accumulates over eigenvalues is not comparable between them. Every statement about least squares in the Results is scoped to one configuration and names it. The statement that least squares cannot deliver the unitary current is a statement about the four-parameter configuration.

Parameter estimation and the two anchors

Inference in each grid cell is two-stage. First, the ensemble is partitioned into groups of consecutive recordings and a separate maximum-likelihood estimate (MLE) is computed per group, producing a cloud of estimates whose scatter is the empirical parameter covariance. The partition is by position and any remainder is dropped, which costs nothing here because the recordings are independent and identically distributed and 10^4 divides by both group sizes used. The optimiser is Gauss–Newton with Levenberg damping, warm-started at the simulation truth. The script sets the damping schedule (initial damping 10, growth factor 3, maximum damping 10^{10}) and three limits: at most 100 iterations, a relative gradient tolerance of 10^{-6} and a value tolerance of 10^{-10}. What stops most fits, however, is a fourth criterion the script cannot reach. The optimiser tests the Newton decrement $\frac{1}{2}\, g^{\mathsf{T}} H^{-1} g$ first, requiring it below 10^{-8} for the first ten iterations and below 10^{-4} after that; the relaxation is there because past that point a flat likelihood is being chased below its own noise floor. The three script-set limits act as fallbacks behind it, and the gradient-norm one is taken relative to the norm of the warm start rather than of the current θ, so that a parameter running away along a flat direction cannot inflate its own tolerance and be recorded as converged. The Newton-decrement constants are compiled in. Two group sizes were used, 10 and 100; with $10{,}000$ recordings these give $1{,}000$ groups of 10 and 100 groups of 100. Second, a single joint MLE is computed over all replicates at once (one group of size n_{sim}), giving the pooled estimate θ_{pool}; its near-zero gradient certifies that it is a stationary point of the pooled likelihood.

A group whose refit fails outright is dropped from the cloud, so a cloud can hold fewer estimates than the partition has groups, and its size is conditional on convergence. Every fit that did complete carries its convergence status, decided by the criteria above, into the per-group run log. The figures in the body do not filter on that status; the supplements that restrict themselves to converged fits say so.

Warm-starting the per-group fits at the truth is deliberate. It isolates the error of the likelihood from the error of the optimiser, which is what this paper studies; it would be the wrong choice for a study of an estimator, which this is not.

The two anchors, the simulation truth θ_{sim} and the pooled fit θ_{pool}, are distinct points because the Gaussian macroscopic likelihood is misspecified with respect to the exact simulator. Their difference $\theta_{\mathrm{pool}} - \theta_{\mathrm{sim}}$ is the misspecification bias. Every diagnostic below is computed at both anchors, and the two answer different questions: at θ_{pool} the score is zero by construction so bias washes out, whereas at θ_{sim} the bias is visible but the Fisher information can be indefinite away from the maximum. Each figure states which anchor it uses.

Diagnostics and the Gaussian-Fisher anchor

Faithfulness of each approximate likelihood to the simulator is measured with likelihood-level diagnostics: standardized residuals, the score (the gradient of the log-likelihood), and an information-distortion matrix that compares the covariance J of the score across replicates with the model's own Fisher information. The definitions, sign conventions, and thresholds of these diagnostics are owned by the Diagnostics section and its metric registry; here we describe only how they were computed.

Every diagnostic reported in this paper is anchored on the analytic Gaussian Fisher information $\bar{G} = \sum_t \mathrm{GFI}_t$, summed over intervals. This is the anchor the Diagnostics section writes $\mathbf{H}$, that is $\mathbf{H} = \bar{G} = \sum_t \mathrm{GFI}_t$. Its per-interval term is

$$\mathrm{GFI}_t = \frac{(\partial_\theta \bar{y}_t)(\partial_\theta \bar{y}_t)^\top}{\sigma_t^2} + \frac{(\partial_\theta \sigma_t^2)(\partial_\theta \sigma_t^2)^\top}{2\,\sigma_t^4},$$

which follows from the analytic derivative alone and needs no extra likelihood passes. It is positive semidefinite by construction, so no anchor used here can be indefinite.

A second Fisher construction exists in the codebase and was used in an earlier battery: a central difference of the analytic score, with per-coordinate step $h_i = h_{\mathrm{rel}} \max(|\theta_i|, 1)$ at $h_{\mathrm{rel}} = 10^{-5}$, symmetrised as $(F_{\mathrm{num}} + F_{\mathrm{num}}^\top)/2$. Because the score itself is obtained by automatic differentiation, this is a single numerical derivative rather than two. That battery (data directory `433ed13`) is retained in the deposit as the demonstration that the two constructions target the same information, an agreement the time-resolved figures exhibit per interval. No number reported in this paper is taken from it.

Comparing the score covariance J against the Fisher information is the sandwich, or information-matrix-equality, diagnostic (***White, 1982***): equality $\mathbf{H} = J$ holds only when the likelihood is correctly specified, and the extent to which the two differ is what the distortion matrix quantifies. The comparison has a further premise on the least-squares branch, where $\bar{G}$ presumes homoscedastic residuals; the Diagnostics section states that premise where the distortion is defined.

Uncertainty on every reported statistic comes from a nonparametric bootstrap over groups, reported as percentile intervals at the 2.5, 16, 50, 84 and 97.5% quantiles. Despite the internal column names, no probit transform is applied; the intervals are ordinary empirical percentiles, verified equal to the R quantile of type 6. The diagnostic battery used 100 bootstrap replicates and a maximum lag of 10 for the score autocorrelation, and the empirical-versus-theoretical distortion capstone used 200. The per-group estimate cloud is deposited raw. The program does run a group bootstrap of it, with a floor of 10 groups below which it falls back to a single unresampled pass, but that result is held in memory and never written: the cloud's only output file is the per-group run log, one row per group and component. Every interval quoted from the cloud in this paper is therefore computed downstream in the figure code, from the raw estimates.

Regime grid and reduction

The acquisition interval was swept over seven levels of Δ/τ, namely 1, 0.5, 0.2, 0.1, 0.05, 0.02 and 0.01, realised as 500, 250, 100, 50, 25, 10 and 5 raw samples averaged per interval, that is step durations of 10, 5, 2, 1, 0.5, 0.2 and 0.1 ms, giving 10, 20, 50, 100, 200, 500 and 1,000 steps in the record. The total record duration is held fixed at 100 ms across the sweep, so the interval is varied without varying the amount of data. Within each record the pre-pulse, agonist and post-pulse segments hold a fixed ratio of $20\,{:}\,40\,{:}\,40$ percent of the steps.

The noise axis needs its conversion stated once, because the quantity the figures plot and the quantity the engine consumes differ by a fixed factor. The engine parameter is `Current_Noise`, a white-noise power spectral density in units of $g^2\,\mathrm{s}$, and it reaches the likelihood only through the variance of one recorded point, $e_t = \mathrm{Current\_Noise}/\Delta_t$, with no correlation time of its own. The sweep is declared by a label, and the dispatchers pair each label with an engine value by a hand-written case statement, so that $\mathrm{label} = 1000\,\mathrm{Current\_Noise}$ exactly. The figure label 0.1 is $\mathrm{Current\_Noise} = 10^{-4}$. The dimensionless noise is

$$\tilde{S} = \frac{\mathrm{Current\_Noise}\; k_{\mathrm{off}}}{i^2},$$

with i the unitary current, so $\tilde{S}$ is the instrumental noise accumulated over one channel time constant $\tau = 1/k_{\mathrm{off}}$, measured against the single-channel signal power. At the run values $k_{\mathrm{off}} = 100\ \mathrm{s}^{-1}$ and $i = g = 1$, this gives

$$\tilde{S} = \frac{\mathrm{label}}{10},$$

so the label is ten times the dimensionless noise it is named after. The factor of ten is the reference sampling interval $\Delta = 0.1\,\tau$ folded into the hardcoded divisor, and it is a naming artefact rather than a design choice; the simulations consumed `Current_Noise` and are unaffected by it. Every stored label is left as written, because those strings tie each figure to the cell that produced it. The figures are drawn in label units and their axes say so; wherever an absolute noise level appears in prose we quote $\tilde{S}$. The reference cell used throughout, label 0.1, is $\tilde{S} = 0.01$, and the label 10 cell is $\tilde{S} = 1$, the level at which the instrumental noise over one channel time constant equals the unitary current.

The mean channel number N_{ch} and the noise level are the other two map axes. The results reported here are read from three data directories, each named by the git commit hash of the build that wrote it: `1c2ae6f`, `87889e6` and `0ffbda7`, searched in that order with the first hit winning. Every cell they contribute to a body figure is at $n_{\mathrm{sim}} = 10^4$, which the figure code enforces by matching `nsim_10000` in the filename, so a cell run at a smaller ensemble cannot enter a map silently. The noise axis is a fixed ladder of twelve labels, and every cell of every algorithm sits on one of them:

$$\mathrm{label} \in \{0.05,\ 0.1,\ 0.2,\ 0.5,\ 1,\ 10,\ 10^2,\ 10^3,\ 10^4,\ 10^5,\ 10^6,\ 10^7\}, \tag{20}$$

which in the dimensionless noise $\tilde{S}$ of this section is one tenth of each. The ladder is denser below one, where the crossover the map measures sits, and by decades above it. Across the three directories, `IR` covers $N_{\mathrm{ch}} \in \{5, 10, 20, 50, 100, 200, 500, 1000, 2000, 5000, 10000\}$ and all twelve labels; NR covers $N_{\mathrm{ch}} \in \{10, 100, 1000, 10000\}$ and R covers those and additionally 5, 20 and 50 at label 0.1, with the highest labels run only at the larger channel counts and reaching 10^7 at $N_{\mathrm{ch}} = 10^4$; MR and VR cover the same channel numbers at labels $\{0.1, 1, 10\}$; and the least-squares arm covers 30 cells on $N_{\mathrm{ch}} \in \{10, 100, 1000, 10000\}$, with labels from 0.1 up to 10^4, 10^5, 10^6 and 10^7 respectively. The label range is deliberately not rectangular: the highest noise labels were run only at the larger channel numbers, where the crossover the map measures sits. The figure code auto-detects which cells exist on disk at render time and prints the detected grid into the render log, so the cell set behind each figure is recoverable from that log.

Each grid cell is reduced through five stages: the per-group MLE cloud; the pooled joint fit θ_{pool}; the score at both anchors, whose across-replicate covariance is J and whose per-interval Gaussian terms sum to the anchor $\bar{G}$; the diagnostic battery at each anchor; and the empirical-versus-theoretical distortion capstone, anchored at θ_{pool}. Each cell writes five comma-separated-value families (the estimate cloud, the pooled fit, the two anchor batteries, and the capstone). The scripts number these stages one, two, four, five and six; the gap at three is the numerical-Fisher stage, which the Gaussian anchor removes and which the numbering keeps visible. On the grid-production lane the cloud stage and the capstone are dropped, because no map reads them there; the cells that need a cloud are produced by the separate script that keeps that stage.

Reproducibility and computational environment

The engine is `macro_dr`, built as C++20, and each analysis is a script in a small typed domain-specific language run in a single process (***Moffatt, 2025***). Provenance is self-documenting at three levels: on every invocation the binary snapshots the assembled script and its full argument list into a run ledger; every analysis output file stamps the build's short git commit hash as its first row; and the data directories are named by that hash, so results from different code versions can never overwrite one another and multi-commit provenance is documented per file.

The production batteries ran under SLURM at 32 CPUs per task and 48 GB of memory (96 GB for the parameter-recovery lane), with the linear-algebra back end pinned to a single thread so that OpenMP parallelises the per-simulation and bootstrap loops. The environment variable `MACRODR_AXIS_SERIAL=1` is load-bearing: it serialises the internal grid-combination loop, without which memory grows with the number of concurrent combinations and the jobs exhaust memory.

The one reproducibility limit worth stating twice is the simulation seed. Because the seed was the random sentinel and the resolved value was not recorded, the exact deposited ensembles cannot be regenerated bit for bit; a rerun yields a statistically equivalent, numerically different ensemble. All resampling that operates on a fixed dataset (the bootstraps, the group partitioning) is deterministic and reproducible given the data. Deposition of the data and code, with pinned versions, is stated in the Data and code availability section.

Data and code availability

All code needed to reproduce the study is openly available. The analysis pipeline that generates the figures, together with the simulation configurations that define every run, is deposited in a public repository (`macroir-validity`) and archived at Zenodo, with the DOI issued on acceptance. The simulation and inference engine is the `macro_dr` software ***Moffatt*** (***2025***), referenced at the pinned version that produced the deposited data; each output file records the engine commit hash that generated it. No experimental data were used: every dataset in the paper is simulated, and is regenerable from the deposited configurations.

A separate library, `macroir`, carries the boundary-conditioned likelihood, its score and its Gaussian Fisher information out of the engine and into a small self-contained C++ core, with an R package and a Python package over it. The scope of that library is narrower than the paper: it implements the boundary-conditioned member, the exact simulator and a Gauss-Newton fit, and it does not implement least squares, the open-loop member or the recursive member on instantaneous samples, so the comparisons reported here are reproducible only from `macro_dr`. The core was transcribed from the engine and checked against it on eight cells, covering four kinetic schemes, three acquisition intervals spanning a factor of a hundred, a protocol whose concentration jumps split sampling intervals, and a recording with a gap, agreeing on the total log-likelihood to between 10^{-16} and 6×10^{-15} relative and on the score to between 8×10^{-14} and 1.4×10^{-9}. The two language packages are bindings over that one core rather than independent reimplementations, and they were run against it once on a single cell, by `tools/cross_language_check.py`, which is the evidence for the statement that the three interfaces return the same numbers.

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Author contributions

L.M. conceived the study, developed the theory, implemented the algorithms, designed and ran the simulations, analysed the data, and wrote the manuscript.

Competing interests

The author declares no competing interests.

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Appendix 1

The interval update, member by member

The Theory section gives the cycle for the boundary-conditioned member and names the two places where the others differ. This appendix writes every member out in one template, so that each is fixed by three switches on a single set of equations.

Objects computed once per interval

All members share the quantities built from the rate matrix $\mathbf{Q}$ and the recording filter for a window of length Δ: the transition matrix $\mathbf{P}(\Delta)$; the boundary-conditioned single-channel moments $\overline{\Gamma}_{i_0\to i_t}$ and $\overline{V}_{i_0\to i_t}$; the start-conditioned mean conductance $\bar{\boldsymbol{\gamma}}_0$ of Eq. 6; the start-conditioned interval variance $\boldsymbol{\sigma}^2_{\bar{\gamma}_0}$; and, for the members that ignore the acquisition averaging, the instantaneous per-state conductance vector $\boldsymbol{\gamma}$. Two matrices collect the boundary weights,

$$G_{i_0 i_t} = \overline{\Gamma}_{i_0\to i_t} P_{i_0\to i_t}(\Delta), \qquad H_{i_0 i_t} = \overline{\Gamma}^2_{i_0\to i_t} P_{i_0\to i_t}(\Delta). \tag{A1}$$

The tilde operator

The tilde is a lift, a modulation and a collapse. An object carried on the K states is lifted to the K^2 boundary pairs (i_0, i_t), multiplied there by the interval weights $\overline{\Gamma}$, $\overline{V}$ and $\mathbf{P}(\Delta)$, and summed back down over the pair index. Its two specializations are the ones the likelihood needs. The bilinear tilde is the scalar of Eq. 11, which contracts the boundary covariance twice. The vector tilde is the gain,

$$\mathbf{g} = \widetilde{\boldsymbol{\gamma}^\top \boldsymbol{\Sigma}} = \mathbf{P}(\Delta)^\top \big(\boldsymbol{\Sigma}_0 - \mathrm{diag}(\boldsymbol{\mu}_0)\big)\bar{\boldsymbol{\gamma}}_0 + \mathbf{G}^\top \boldsymbol{\mu}_0, \tag{A2}$$

which contracts it once and leaves a K-vector, the cross-covariance between the occupancy at the end of the window and the interval current. Both are evaluated in $K \times K$ arithmetic; the $K^2 \times K^2$ boundary covariance is never built.

One cycle, three switches

Every member on the lattice runs the same four steps on each window: propagate the occupancy Gaussian (Eqs. 7 and 8), predict the observation (Eqs. 9 and 10), update by the scalar Kalman correction (Eqs. 12 and 13), and either carry the posterior to the next window or discard it. Three switches fix which member is being run.

Switch 1: the window, $\mathrm{av} \in \{0, 1, 2\}$.

It selects the conductance object, and with it the second term of the predictive variance and the gain. At $\mathrm{av} = 0$ the observation is treated as an instantaneous sample taken at the centre of the window: the occupancy is propagated there first, by $\mathbf{P}(\Delta/2)$, the conductance is $\boldsymbol{\gamma}$, and, with $\boldsymbol{\Sigma}$ read in that mid-window frame,

$$\widetilde{\boldsymbol{\gamma}^\top \boldsymbol{\Sigma} \boldsymbol{\gamma}} \to \boldsymbol{\gamma}^\top \boldsymbol{\Sigma} \boldsymbol{\gamma}, \qquad \mathbf{g} \to \mathbf{P}(\Delta/2)^\top \boldsymbol{\Sigma}\,\boldsymbol{\gamma}. \tag{A3}$$

At $\mathrm{av} = 1$ the same two expressions hold with $\bar{\boldsymbol{\gamma}}_0$ in place of $\boldsymbol{\gamma}$ and the full $\mathbf{P}(\Delta)$ in the gain, since the conductance is conditioned on the state at the window's start and the whole window remains to be crossed; the contraction is still an ordinary quadratic form on the K states. The trailing propagator in the gain is not decoration. It carries the update from the frame in which the observation was formed to the frame the next window starts in, which is the end of the current window, so it is the remaining half at $\mathrm{av} = 0$ and the whole window at $\mathrm{av} = 1$. Omitting it, or using the full window where half is called for, leaves the rank-one down-date in the wrong frame, and the symptom is specific and was observed: a spike in the distortion-induced bias of the recursive instantaneous member at long intervals and large channel counts. conductance is conditioned on both endpoints and the two expressions become the tildes, Eq. 11 and Eq. A2. The step from $\mathrm{av} = 1$ to $\mathrm{av} = 2$ therefore does two things at once: it replaces $\boldsymbol{\Sigma}_0$ by $\boldsymbol{\Sigma}_0 - \mathrm{diag}(\boldsymbol{\mu}_0)$ and adds the boundary term ($\boldsymbol{\mu}_0^\top \mathbf{H1}$ in the variance, $\mathbf{G}^\top \boldsymbol{\mu}_0$ in the gain). The subtracted diagonal is not a second approximation: the within-channel part it removes is already inside the boundary term, and dropping the subtraction would count it twice.

Switch 2: recursion.

A recursive member carries $(\boldsymbol{\mu}^{\mathrm{post}}, \boldsymbol{\Sigma}^{\mathrm{post}})$ into the next window. A non-recursive one discards it and propagates the next window's predictive from the initial condition, so its log-likelihood factorizes over windows. The switch changes no formula above; it changes only what is fed to the next one.

Switch 3: the interval variance, total or residual.

It selects the third term of Eq. 10, which is present only when the acquisition averaging is modelled at all $(\mathrm{av} \geq 1)$:

$$\left(\boldsymbol{\sigma}^2_{\bar{\gamma}_0}\right)_{i_0} = \begin{cases} \sum_{i_t} P_{i_0\to i_t}(\Delta)\left[\overline{V}_{i_0\to i_t} + \left(\overline{\Gamma}_{i_0\to i_t} - (\bar{\gamma}_0)_{i_0}\right)^2\right] & \text{total,} \\ \sum_{i_t} P_{i_0\to i_t}(\Delta)\,\overline{V}_{i_0\to i_t} & \text{residual.} \end{cases} \tag{A4}$$

The two differ by the spread of the interval mean across end states, which is Eq. 14. The switch is free only at $\mathrm{av} = 1$. At $\mathrm{av} = 2$ the residual form is forced whatever the flag says, for the same reason the diagonal is subtracted in switch 1: the between-endpoint spread is already carried by the boundary term, so the total form would count it twice.

The members

Member	av	recursive	interval variance	what it is
`NR`	0	no	—	instantaneous samples, windows independent
`R`	0	yes	—	instantaneous samples, posterior carried forward
`INR`	1	no	total	`NR` with the window average restored
`MR`	1	yes	total	`R` with the window average restored
`VR`	1	yes	residual	`MR` with switch 3 flipped, gain unchanged
`IR`	2	yes	residual (forced)	`VR` with the boundary terms restored

Appendix 1—table 1. The six members of the lattice as settings of the three switches. Reading down the last column, each row differs from an earlier one in exactly one switch, which is what makes the ladder measurable one step at a time. `IR` is MacroIR; `INR` is the MacroINR of ***Moffatt and Pierdominici-Sottile*** (***2025***). The data keys under which each is dispatched are in Table 2.

One consequence of the two switches acting together is worth stating, because it is easy to read the table as saying otherwise. `MR` and `IR` predict the *same* total variance. Expanding the total form of Eq. A4 and using $\sum_{i_t} P_{i_0\to i_t}(\Delta) = 1$ and $\sum_{i_t} P_{i_0\to i_t}(\Delta)\overline{\Gamma}_{i_0\to i_t} = (\bar{\gamma}_0)_{i_0}$,

$$\boldsymbol{\mu}_0^\top \boldsymbol{\sigma}^2_{\bar{\gamma}_0}\Big|_{\text{total}} = \boldsymbol{\mu}_0^\top \boldsymbol{\sigma}^2_{\bar{\gamma}_0}\Big|_{\text{residual}} + \boldsymbol{\mu}_0^\top \mathbf{H1} - \bar{\boldsymbol{\gamma}}_0^\top \mathrm{diag}(\boldsymbol{\mu}_0)\,\bar{\boldsymbol{\gamma}}_0, \tag{A5}$$

and the two extra pieces are exactly what switch 1 moves from the third term of Eq. 10 into the second when av goes from 1 to 2. The decompositions differ, the sum does not. What does change between the two members is the gain, and that is the whole of the `MR`-to-`IR` difference.

Two of the six pairs isolate a single algebraic change and are the ones the Results use. Stepping `MR` to `VR` flips switch 3 and touches nothing else, so it measures what the variance form alone is worth. Stepping `VR` to `IR` restores the boundary terms in both the variance and the gain and leaves the variance form where it already was, so it measures what conditioning on the second endpoint alone is worth. The two steps together are the `R`-to-`IR` gap.

Least squares, which is off the lattice

Classical nonlinear least squares has no setting of the three switches. It propagates the mean occupancy alone, $\boldsymbol{\mu}^{\mathrm{prop}} = \mathbf{P}(\Delta)^{\top}\boldsymbol{\mu}_0$, with no covariance and no update, predicts $\bar{y}^{\mathrm{pred}}$ from Eq. 9, and assigns one constant variance to every residual, fitted as the residual sum of squares over the number of samples. Neither term of Eq. 10 that depends on the gating survives, and there is no gain, so no occupancy object is ever conditioned on the data. It is the anchor of the comparison rather than a member of the family.

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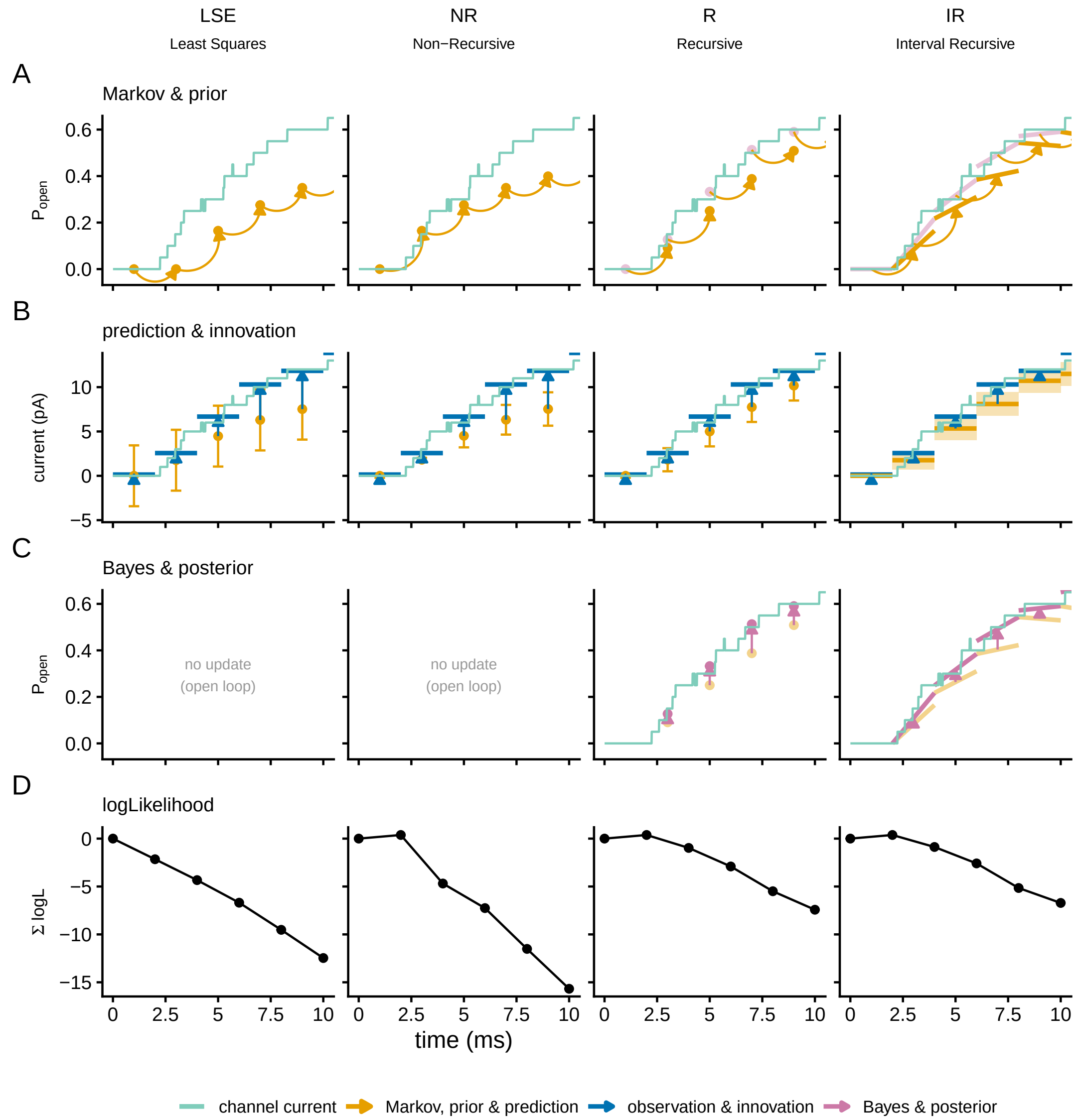

Figure 1—figure supplement 1. The same filter cycle over the first ten milliseconds of the recording, so that the concentration jump and the onset of gating are inside the window.

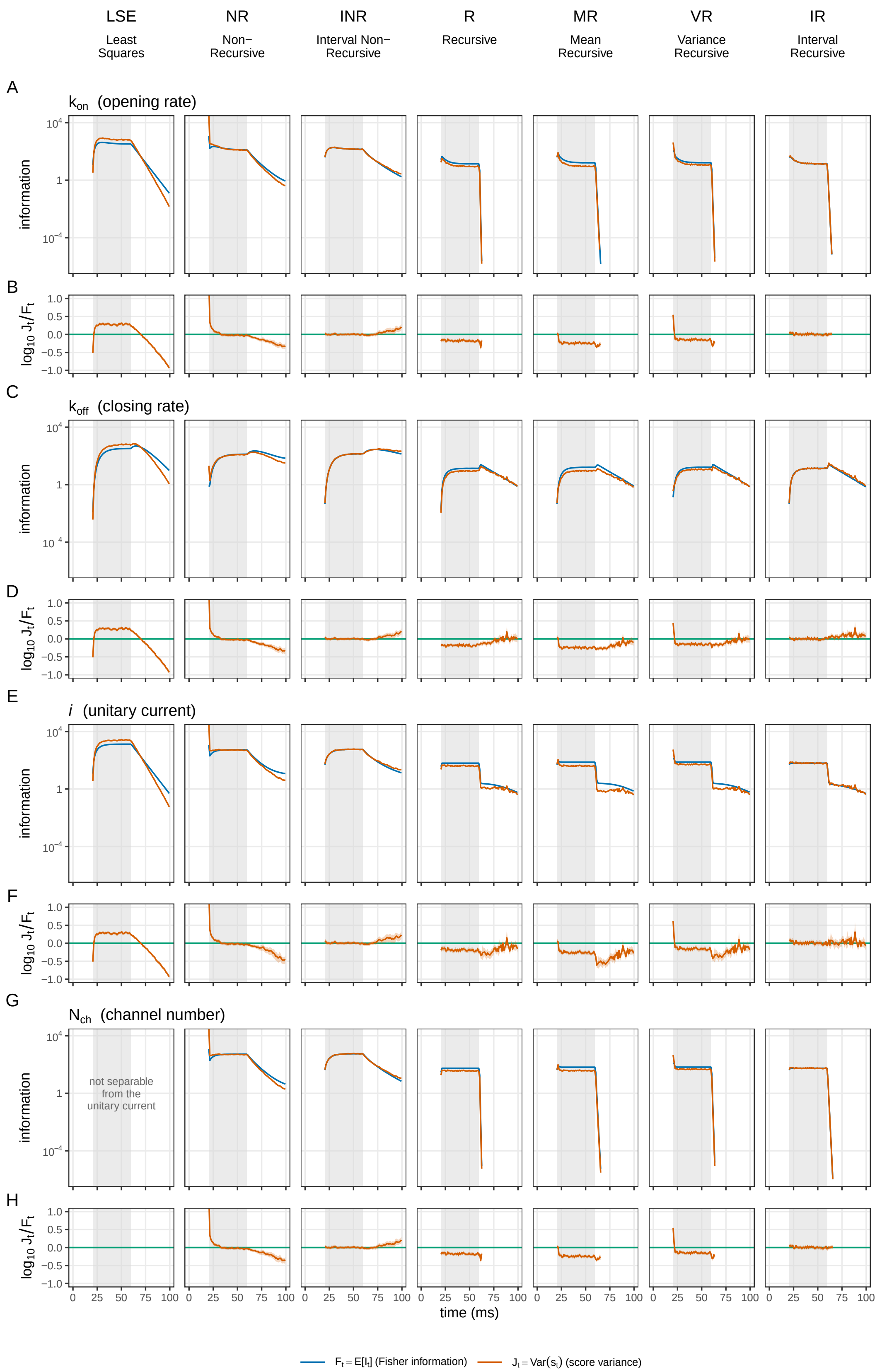

Figure 3—figure supplement 1. Per-step information against score variance, for every member and all four identified parameters. All seven members across the columns and the four parameters down the page as bands of two rows: k_{on} (A, B), k_{off} (C, D), the unitary current (E, F) and N_{ch} (G, H). The upper row of each band carries the per-interval Fisher information and the score variance as magnitudes, which lie on each other where the member is calibrated at that interval; the lower row is $\log_{10}(J_t/F_t)$ with its 95% bootstrap interval, zero when they do. Per-interval calibration is not monotone along the cost ladder: on k_{off} the median reads $+0.184$ for LSE, -0.038 for NR,

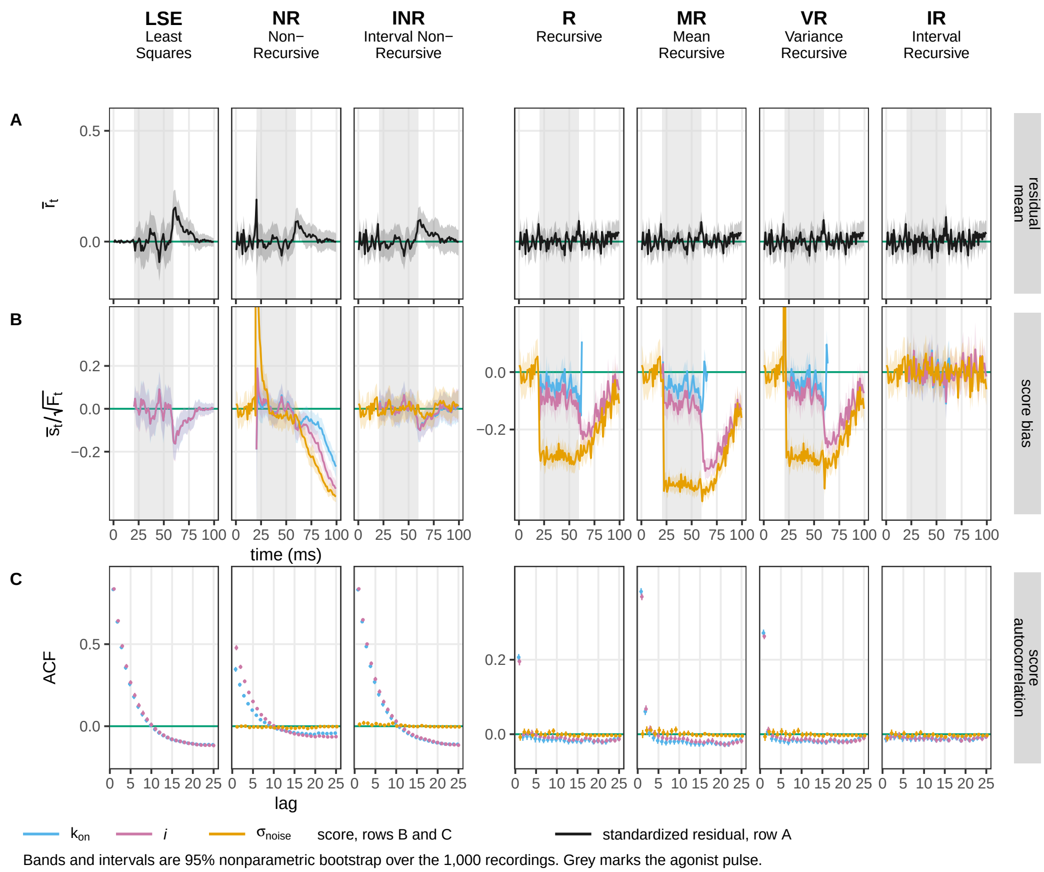

Figure 3—figure supplement 2. The checks this figure has no room for. All seven members at the same cell, carrying (A) the mean standardized residual, the one of the three properties of r_t not drawn above; (B) the score bias for the three parameters not coloured above, k_{on}, the unitary current and the instrumental noise level; and (C) their score autocorrelation. The first residual moment separates nobody, which is what makes the other two properties the informative ones. In (B) the ordering repeats the one in figure supplement 2: counting informative intervals whose interval excludes zero against a chance level of 0.05, IR reads 0.067, 0.050 and 0.050, INR 0.100, 0.100 and 0.000, and the three recursive members between 0.409 and 0.975. The noise level is the control direction: it is the only parameter measured before the agonist arrives, and its score is white even in the members whose kinetic directions are not, so the memory that carries the accumulated information error lives in the gating directions and not in the instrumental one. Least squares carries no noise curve because its noise level is a plug-in from the same record rather than a free direction, so its score there vanishes identically, which is the same fact as its mean squared residual being pinned at one.

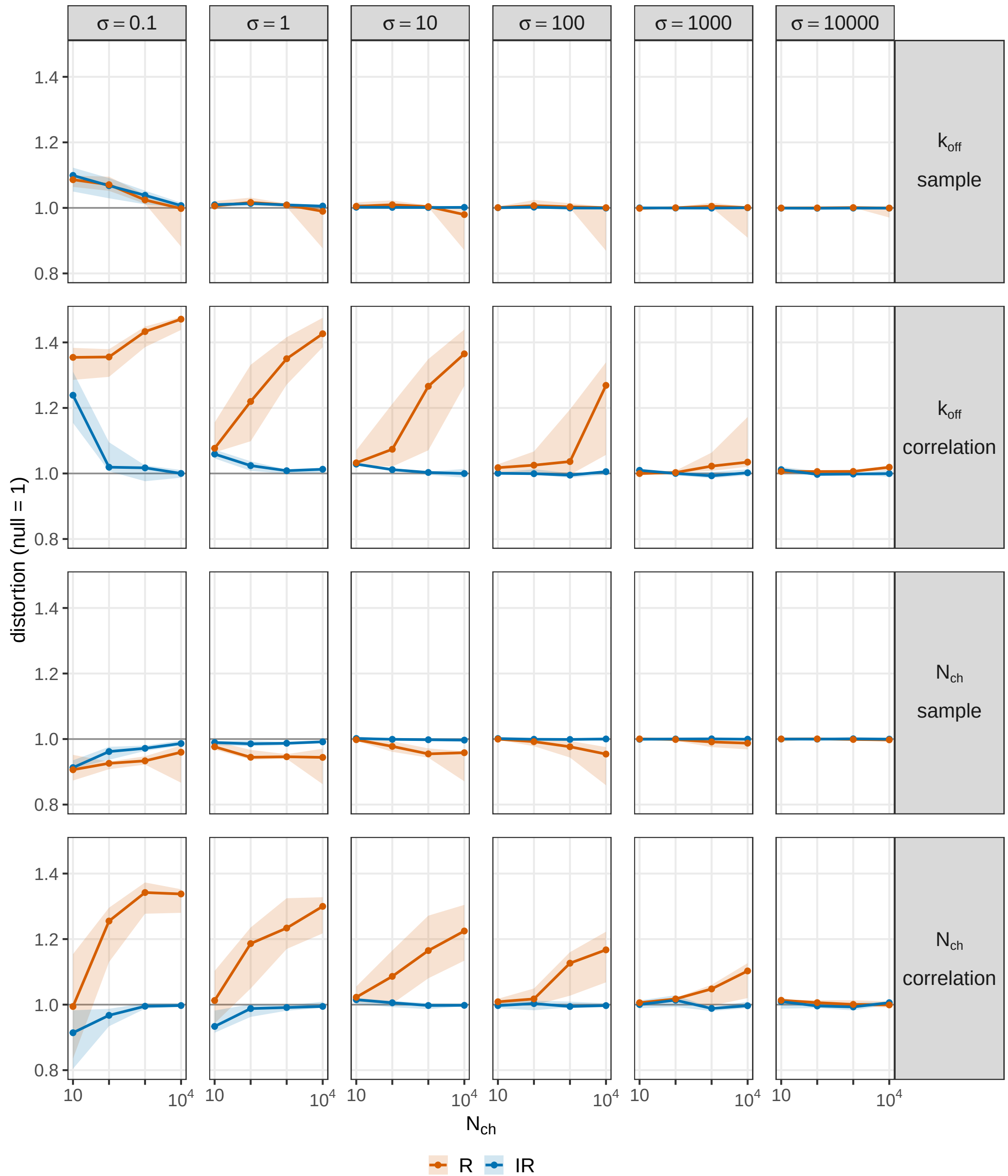

Figure 4—figure supplement 1. The distortion decomposed into a per-sample part and a correlation part, against channel count and resolved by instrumental noise. The per-sample parts of the two recursive members coincide and go to one as channels are added, while the correlation parts separate.

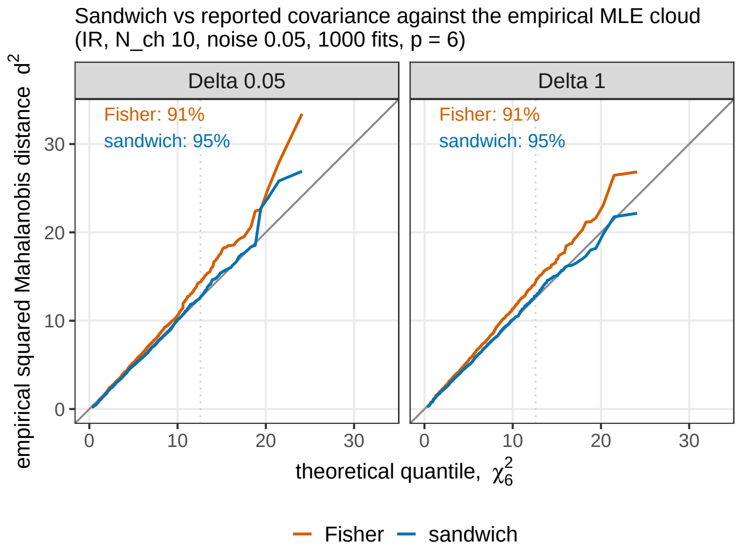

Figure 4—figure supplement 2. The sandwich prediction against the empirical multivariate distribution of the estimates, as a Mahalanobis quantile-quantile plot with a χ^2 reference. The reported Fisher covariance under-covers the nominal 95% region and the sandwich covers it.